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1. Ten Simple Rules for Organizing and Running a Successful Intensive Two-Week Course

Gunnar Blohm, Paul Schrater, Konrad P. Körding

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Intensive summer schools often provide strong students with career-changing impact, teaching them the art of the trade, letting them understand the logical underbelly of a field, and connecting them with an elite circle of peers and field leaders. Indeed, many professors attribute considerable aspects of their growth as a scientist to such schools. Such summer schools are an essential service to the community. A well-run summer school combines many of the aspects that jointly define students overall success. Eight years of organizing the annual two-week Computational Sensory-Motor Neuroscience (CoSMo, http://www.compneurosci.com/CoSMo) summer school has allowed us to experiment with different approaches and evaluate teaching outcomes, and we have seen rather clear patterns. Many new schools are started each year, only some move on to ongoing success, and the vast majority take a while until they reach very good ratings (our ratings are still increasing but approaching 10/10). Focusing on the student experience, we present a set of 10 simple rules to help you organize better summer schools that are more useful to students.We organize summer schools despite the considerable cost in terms of money and effort associated with them. We do this because we want to improve the field. Understanding the students and the change we want to effect is essential to being effective. For example, CoSMo mostly serves the objective of improving the way the field handles data and computation. Everything else derives from this objective. Teaching objectives should thus be defined by answering the following questions: What is missing in the field? What are the educational bottlenecks in the field? Where would we like the field to move? We should think of summer schools not as a generic way of teaching students but a specific way of effecting the change the field needs. Organizing a summer school is a way of leading the change you want to see.Knowing the students also implies not overloading them. The learning experience is far better when students learn material that they can realistically acquire than if an overly ambitious program instead teaches the students a little bit of many things without giving them important new skills. A summer school is no replacement for an education in science. It is a boost and a little bit of a push into the right direction.Crucial to the success of a teaching objective is the recruitment of pedagogically outstanding lecturers. Those individuals do not have to be the biggest names in the field; rather, they should be good teachers and should have the required hands-on practical knowledge to tutor participants. This said, students are attracted by faculty stars, so consider a healthy balance between fame and pedagogy. It is often useful to recruit additional teaching assistants if necessary to provide a high teacher-to-student ratio where each student can receive individualized attention. Often there are also experts among the participants, and one can take advantage of impromptu peer mentoring approaches. Fewer educators are better than many so that a coherent teaching framework can be applied over one to two days for each lecturer team. If multiple lecturers coteach a topic, true coteaching can be tremendously beneficial (but requires serious initial organization). We found that reducing the number of lecturers was almost universally associated with improved perceived quality of education ratings.Summer school participants usually come with a broad variety of educational backgrounds and can be at different career stages. It is thus important to design the summer school teaching content in a way that provides enough introductory material in order to equalize knowledge before more specialized or

in-depth topics are covered. Our experience shows that about half the content of the summer school should be structured to convey the basics of a research field. These basics should ideally be taught by a small team of instructors (e.g., the way the three organizers of CoSMo do this) to ensure maximum coherence and coordination of content. Indeed, the content should be selected with at least two criteria in mind: it should level the ground for the following guest lecturers, who typically cover more specialized, in-depth content, and it should provide a coherent but critical perspective of current approaches in the field, including philosophical considerations.A perennial problem when teaching the basics is how to provide the more advanced students with good learning opportunities while helping weaker students advance. We found that keeping tutorials multilevel can be good at solving this issue. First, the advanced students can be assigned to help the less advanced ones at certain times. Second, it is often possible to provide an extra-hard problem in the same tutorial content. When we teach, we often have the hard problems at the bottom of the slide and the main ones in the middle.The aspect that sets a summer school apart from university-based teaching, lectures at conferences, and reading material is that due to the targeted topic and small number of students at a summer school, we can teach skills instead of material. Of course, some lecturing is important to introduce certain concepts and background. In a tutorial, students can learn how to solve problems in a way that will almost immediately allow them to replicate those skills back in their own lab. We found that in computational neuroscience, assigning more than half of the time to tutorials considerably improves what students learn.We also found that switching to a microlecture-tutorial format considerably improved outcomes. We teach a concept for a few minutes, and then for about the next 10 minutes, students immediately use what they just learned to solve a concrete problem that they could encounter in their own lab. This immediate link between knowledge and its application makes it both easier to attend the lecture because it will become relevant knowledge in 5 minutes. It is also easier for students not to get lost. The application makes it easy for them to see what they have not understood. Learning outcomes are optimized through active learning. Problem-based learning tutorials are best because they require critical thinking and the development of logical solutions. They implicitly force participants to ask the right questions, identify assumptions, and make hypotheses explicit. This can be a slow process, but it ensures a profound and practical understanding. Also, whenever possible, choose a single data set or problem for participants to work on from different angles in order to reduce introduction time. Thus, often when it comes to selecting course content, less is more.The venue should also be adapted to the teaching objectives. Interactive classrooms with round tables, lots of whiteboard space, breakout rooms, and audio-video equipment in many cases greatly increases teaching efficiency. It is truly helpful if the more advanced students can help those who are less advanced understand the material. An open design (e.g., with round tables) makes it easy for students to interact with many peers.We also found that switching activities is important to keep up participants' energy and attention span. Frequent switching between lectures and tutorials, switching between lecturers, switching groups for tutorial work, taking breaks, and interjecting special activities (e.g., sports, a special lecture, an open discussion, telling an anecdote) all help to increase the teaching outcomes.Tutorials allow students to work on only small problems. A small group project (three to five group members are ideal) that spans the entire duration of the summer school allows a more in-depth treatment of a selected topic. It also fosters more creativity, additional individualized teaching opportunities, and opportunities for participants to more deeply network with other students with whom they have the most in common. For most summer schools that we know of, combining these two timescales of learning is perceived as productive.Importantly, for group projects to be truly productive, it is essential that they are guided and tutored in the right way. Students need considerable guidance at multiple points along the way. First, it greatly helps if the group generation process is guided. Students should build groups that cover complementary skills while having common interests. Group parity should also be targeted because it leads to better group dynamics. Second,

it is essential that group projects are tutored, but in a way that lets the participants steer them. The students can often learn a lot from learning why a given project is not a good group project. Third, it needs a final set of maximally public presentations so that the students have the sense that they are working toward an important goal. To maximize this effect, we convinced those at one of our field's main annual conferences to automatically accept the best paper presented at our summer school in a year. Moreover, CoSMo summer projects regularly end up being published as scientific papers in established journals.The pace for the entire summer school, and in particular the group projects, is set in the first two days or so. Long work hours (e.g., 9:00 a.m. to 11:00 p.m.) create group cohesion (through “suffering” together) and promote a community spirit that is fertile for learning, ideas, and collaboration. This atmosphere can be enhanced by providing an immersive learning environment where lecturers and tutors are available at all times. Make sure enough time is available for group projects.Teaching materials, including lectures, tutorials, solutions, and additional resources, should be publicly shared, for example, on a wiki site (e.g., http://compneurosci.com/wiki/index.php/CoSMo) or www.OSF.io. This crucial component has many benefits. It is, of course, useful for students to always have access to all materials. It is also useful for instructors to see what previous materials have been covered. More important, open access of materials is immensely useful for the community and in high demand. It is one way to extend the impact of a summer school far beyond the limited group of those attending. Other professors will use tutorial and lecture materials for their own classes. Researchers have a means to learn new approaches on their own, which can tremendously accelerate a field's research endeavor, especially when example code from tutorials is advanced enough that it can be directly applied to research questions. Finally, a public written trace is also useful for summer school lecturers to keep a good institutional memory. It is critical to remember over the years what has been taught, what has worked, and what needs to be improved (see also rule 10).In fact, many of the materials developed for CoSMo ended up being used more broadly across the field. For example, a tutorial aimed at teaching multivoxel pattern analysis ended up being a crucial resource for teaching in the entire field (http://www.cosmomvpa.org/). In many ways, the online materials are a continuation of the summer school itself.A summer school is a unique networking event that can build lifelong professional relationships. It is thus important to promote networking and have enough unstructured time available, even if that means prolonged lunch and dinner breaks. This can be done in many ways: through creating mailing lists, blogs, and forums; using social media; and organizing group events, (for example, an outing after the first week of the school, such as sightseeing or sports activities). Networking should not be limited to participants but should include all lecturers.Our alumni organize get-togethers at the big national conferences. They share news of personal progress on the Facebook page for the course session they attended. They hire one another once they become professors. And they often end up collaborating with one another later in their careers. Keeping the network in place after a summer school is crucial to the success of the participants.We found a simple yet critical way to increase the effectiveness of the summer school. A problem for probably most of our students is access to research supervisors in their home universities. The students may have a supervisor, but that person may not have much time to give to his or her students. In addition, they have a conflict of interest as mentors. Therefore, having one-on-one access at the summer school to their field's leaders is incredibly valuable to these students.We thus ask all students which of the professors they most would like to meet individually. Our guest lecturers coteach, each staying for two days but teaching only for one. The remaining time can then be used for a large number of one-on-one slots. Being able to sign up for one of these meetings, which last for 15 to 30 minutes, is one of the highlights for CoSMo attendees. Indeed, some have told us that this was the highlight of the school. The goal of these one-on-one meetings is to allow participants to establish a personal relationship with established researchers in their field. It allows participants to ask questions in a private setting while having

the full attention of the experts. Conversation topics can range from career advice to feedback on research projects, work-life balance, or recruitment interviews. Other schools achieve the same aims simply by having faculty around for a long time and allocating a lot of time for unstructured networking.Organizing and running a summer school is a lot of work, and the schools themselves are tiring for lecturers, attendees, and the organizers. In order to avoid burnout, maintaining a fun and supportive environment is crucial. Organizers and lecturers need to bring and share positive energy. Positive energy is contagious, and participants draw from the lecturers' energy. Make sure to maintain this positive energy. The emotional side of a summer school is essential. Happy lecturers come again, happy students spread the word and successfully network, and happy organizers will be more willing to go through all that work again next year.Positive energy can be cultivated in a number of ways and should be spread from the beginning. One way is to spend time with your co-organizers and lecturers without students, even if that means less available networking time. It is essential to be there for one another as the teaching team but also essential to see the students as partners in learning. And, quite possibly, we should celebrate a summer school well run. By coteaching and co-organizeing with people we like, the fun will spill over to the participants. It also helps to plan for some fun bonding activities (e.g., sports, games) and make sure there is time for relaxation (e.g., cultural activities, day off, pub night), and not to overwhelm you or the participants.Successful summer schools learn from their mistakes and improve over the years, So ask for feedback from participants. Students often understand what would have helped them. After the school, thier feedback can be followed up with questions asked through Google Forms. The feedback forms should be a mix of rating scales to get participants' overall impressions on what was good and what was not and free-form text for them to provide their feedback about the individual school activities. For example, you can ask separately about lecture versus tutorial content for each teaching component or each lecturer. What was the difficulty level? Did participants feel they really learned something? What improvements could be made?Once feedback has been collected, analyze the ratings and suggestions. Often you need to read between the lines to identify the underlying causes of negative feedback or even to understand constructive suggestions. Always try to improve the summer school. Even after running CoSMo for eight years, we still see room for improvement. A research community is not static, and thinking as well as techniques evolve. Feedback also allows us to identify unmet expectations and make adjustments based on the direction the field is heading in.We thank Pascal Wallisch, Terry Sejnowski, and Adrienne Fairhall for helpful feedback. We also thank all 2011–2018 CoSMo attendees for their constructive (anonymous) feedback that has tremendously helped us shape these 10 simple rules.

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2. New Families of Skewed Higher-Order Kernel Estimators to Solve the BSS/ICA Problem for Multimodal Sources Mixtures

Ahmed N. Jabbar

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This letter suggests two new types of asymmetrical higher-order kernels (HOK) that are generated using the orthogonal polynomials Laguerre (positive or right skew) and Bessel (negative or left skew). These skewed HOK are implemented in the blind source separation/independent component analysis (BSS/ICA) algorithm. The tests for these proposed HOK are accomplished using three scenarios to simulate a real environment using actual sound sources, an environment of mixtures of multimodal fast-changing probability density function (pdf) sources that represent a challenge to the symmetrical HOK, and an environment of an adverse case (near gaussian). The separation is performed by minimizing the mutual information (MI) among the mixed sources. The performance of the skewed kernels is compared to the performance of the

standard kernels such as Epanechnikov, bisquare, trisquare, and gaussian and the performance of the symmetrical HOK generated using the polynomials Chebyshev1, Chebyshev2, Gegenbauer, Jacobi, and Legendre to the tenth order. The gaussian HOK are generated using the Hermite polynomial and the Wand and Schucany procedure. The comparison among the 96 kernels is based on the average intersymbol interference ratio (AISIR) and the time needed to complete the separation. In terms of AISIR, the skewed kernels' performance is better than that of the standard kernels and rivals most of the symmetrical kernels' performance. The importance of these new skewed HOK is manifested in the environment of the multimodal pdf mixtures. In such an environment, the skewed HOK come in first place compared with the symmetrical HOK. These new families can substitute for symmetrical HOKs in such applications.

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3. First Passage Time Memory Lifetimes for Simple, Multistate Synapses: Beyond the Eigenvector Requirement

Terry Elliott

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Models of associative memory with discrete-strength synapses are palimpsests, learning new memories by forgetting old ones. Memory lifetimes can be defined by the mean first passage time (MFPT) for a perceptron's activation to fall below firing threshold. By imposing the condition that the vector of possible strengths available to a synapse is a left eigenvector of the stochastic matrix governing transitions in strength, we previously derived results for MFPTs and first passage time (FPT) distributions in models with simple, multistate synapses. This condition permits jump moments to be computed via a 1-dimensional Fokker-Planck approach. Here, we study memory lifetimes in the absence of this condition. To do so, we must introduce additional variables, including the perceptron activation, that parameterize synaptic configurations, permitting Markovian dynamics in these variables to be formulated. FPT problems in these variables require solving multidimensional partial differential or integral equations. However, the FPT dynamics can be analytically well approximated by focusing on the slowest eigenmode in this higher-dimensional space. We may also obtain a much better approximation by restricting to the two dominant variables in this space, the restriction making numerical methods tractable. Analytical and numerical methods are in excellent agreement with simulation data, validating our methods. These methods prepare the ground for the study of FPT memory lifetimes with complex rather than simple, multistate synapses.

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4. Decoding of Neural Data Using Cohomological Feature Extraction

Erik Rybakken, Nils A. Baas, Benjamin Dunn

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We introduce a novel data-driven approach to discover and decode features in the neural code coming from large population neural recordings with minimal assumptions, using cohomological feature extraction. We apply our approach to neural recordings of mice moving freely in a box, where we find a circular feature. We then observe that the decoded value corresponds well to the head direction of the mouse. Thus, we capture head direction cells and decode the head direction from the neural population activity without having to process the mouse's behavior. Interestingly, the decoded values convey more information about the neural activity than the tracked head direction does, with differences that have some spatial organization. Finally,

we note that the residual population activity, after the head direction has been accounted for, retains some low-dimensional structure that is correlated with the speed of the mouse.

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5. Fixed Points of Competitive Threshold-Linear Networks

Carina Curto, Jesse Geneson, Katherine Morrison

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Threshold-linear networks (TLNs) are models of neural networks that consist of simple, perceptron-like neurons and exhibit nonlinear dynamics determined by the network's connectivity. The fixed points of a TLN, including both stable and unstable equilibria, play a critical role in shaping its emergent dynamics. In this work, we provide two novel characterizations for the set of fixed points of a competitive TLN: the first is in terms of a simple sign condition, while the second relies on the concept of domination. We apply these results to a special family of TLNs, called combinatorial threshold-linear networks (CTLNs), whose connectivity matrices are defined from directed graphs. This leads us to prove a series of graph rules that enable one to determine fixed points of a CTLN by analyzing the underlying graph. In addition, we study larger networks composed of smaller building block subnetworks and prove several theorems relating the fixed points of the full network to those of its components. Our results provide the foundation for a kind of graphical calculus to infer features of the dynamics from a network's connectivity.

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6. Supervised Dimensionality Reduction on Grassmannian for Image Set Recognition

Tianci Liu, Zelin Shi, Yunpeng Liu

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Modeling videos and image sets by linear subspaces has achieved great success in various visual recognition tasks. However, subspaces constructed from visual data are always notoriously embedded in a high-dimensional ambient space, which limits the applicability of existing techniques. This letter explores the possibility of proposing a geometry-aware framework for constructing lower-dimensional subspaces with maximum discriminative power from high-dimensional subspaces in the supervised scenario. In particular, we make use of Riemannian geometry and optimization techniques on matrix manifolds to learn an orthogonal projection, which shows that the learning process can be formulated as an unconstrained optimization problem on a Grassmann manifold. With this natural geometry, any metric on the Grassmann manifold can theoretically be used in our model. Experimental evaluations on several data sets show that our approach results in significantly higher accuracy than other state-of-the-art algorithms.

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7. The Exact VC Dimension of the WiSARD n-Tuple Classifier

Hugo Carneiro, Carlos E. Pedreira, Felipe M. G. França, Priscila M. V. Lima

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The Wilkie, Stonham, and Aleksander recognition device (WiSARD) [Formula: see text]-tuple classifier is a multiclass weightless neural network capable of learning a given pattern in a single step. Its architecture is determined by the number of classes it should discriminate. A target class is represented by a structure

called a discriminator, which is composed of [Formula: see text] RAM nodes, each of them addressed by an [Formula: see text]-tuple. Previous studies were carried out in order to mitigate an important problem of the WiSARD [Formula: see text]-tuple classifier: having its RAM nodes saturated when trained by a large data set. Finding the VC dimension of the WiSARD [Formula: see text]-tuple classifier was one of those studies. Although no exact value was found, tight bounds were discovered. Later, the bleaching technique was proposed as a means to avoid saturation. Recent empirical results with the bleaching extension showed that the WiSARD [Formula: see text]-tuple classifier can achieve high accuracies with low variance in a great range of tasks. Theoretical studies had not been conducted with that extension previously. This work presents the exact VC dimension of the basic two-class WiSARD [Formula: see text]-tuple classifier, which is linearly proportional to the number of RAM nodes belonging to a discriminator, and exponentially to their addressing tuple length, precisely [Formula: see text]. The exact VC dimension of the bleaching extension to the WiSARD [Formula: see text]-tuple classifier, whose value is the same as that of the basic model, is also produced. Such a result confirms that the bleaching technique is indeed an enhancement to the basic WiSARD [Formula: see text]-tuple classifier as it does no harm to the generalization capability of the original paradigm.

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8. Dual Neural Network Method for Solving Multiple Definite Integrals

Haibin Li, Yangtian Li, Shangjie Li

https://doi.org/10.1162/neco_a_01145 | pp. 208-232

This study, which examines a calculation method on the basis of a dual neural network for solving multiple definite integrals, addresses the problems of inefficiency, inaccuracy, and difficulty in finding solutions. First, the method offers a dual neural network method to construct a primitive function of the integral problem; it can approximate the primitive function of any given integrand with any precision. On this basis, a neural network calculation method that can solve multiple definite integrals whose upper and lower bounds are arbitrarily given is obtained with repeated applications of the dual neural network to construction of the primitive function. Example simulations indicate that compared with traditional methods, the proposed algorithm is more efficient and precise in obtaining solutions for multiple integrals with unknown integrand, except for the finite input-output data points. The advantages of the proposed method include the following: (1) integral multiplicity shows no influence and restriction on the employment of the method; (2) only a finite set of known sample points is required without the need to know the exact analytical expression of the integrand; and (3) high calculation accuracy is obtained for multiple definite integrals whose integrand is expressed by sample data points.

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9. Modeling the Correlated Activity of Neural Populations: A Review

Christophe Gardella, Olivier Marre, Thierry Mora

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The principles of neural encoding and computations are inherently collective and usually involve large populations of interacting neurons with highly correlated activities. While theories of neural function have long recognized the importance of collective effects in populations of neurons, only in the past two decades has it become possible to record from many cells simultaneously using advanced experimental techniques with single-spike resolution and to relate these correlations to function and behavior. This review focuses on

the modeling and inference approaches that have been recently developed to describe the correlated spiking activity of populations of neurons. We cover a variety of models describing correlations between pairs of neurons, as well as between larger groups, synchronous or delayed in time, with or without the explicit influence of the stimulus, and including or not latent variables. We discuss the advantages and drawbacks or each method, as well as the computational challenges related to their application to recordings of ever larger populations.

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10. Functional Diversity in the Retina Improves the Population Code

Michael J. Berry, Felix Lebois, Avi Ziskind, Rava Azeredo da Silveira

https://doi.org/10.1162/neco_a_01158 | pp. 270-311

Within a given brain region, individual neurons exhibit a wide variety of different feature selectivities. Here, we investigated the impact of this extensive functional diversity on the population neural code. Our approach was to build optimal decoders to discriminate among stimuli using the spiking output of a real, measured neural population and compare its performance against a matched, homogeneous neural population with the same number of cells and spikes. Analyzing large populations of retinal ganglion cells, we found that the real, heterogeneous population can yield a discrimination error lower than the homogeneous population by several orders of magnitude and consequently can encode much more visual information. This effect increases with population size and with graded degrees of heterogeneity. We complemented these results with an analysis of coding based on the Chernoff distance, as well as derivations of inequalities on coding in certain limits, from which we can conclude that the beneficial effect of heterogeneity occurs over a broad set of conditions. Together, our results indicate that the presence of functional diversity in neural populations can enhance their coding fidelity appreciably. A noteworthy outcome of our study is that this effect can be extremely strong and should be taken into account when investigating design principles for neural circuits.

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11. Equivalence of Equilibrium Propagation and Recurrent Backpropagation

Benjamin Scellier, Yoshua Bengio

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Recurrent backpropagation and equilibrium propagation are supervised learning algorithms for fixed-point recurrent neural networks, which differ in their second phase. In the first phase, both algorithms converge to a fixed point that corresponds to the configuration where the prediction is made. In the second phase, equilibrium propagation relaxes to another nearby fixed point corresponding to smaller prediction error, whereas recurrent backpropagation uses a side network to compute error derivatives iteratively. In this work, we establish a close connection between these two algorithms. We show that at every moment in the second phase, the temporal derivatives of the neural activities in equilibrium propagation are equal to the error derivatives computed iteratively by recurrent backpropagation in the side network. This work shows that it is not required to have a side network for the computation of error derivatives and supports the hypothesis that in biological neural networks, temporal derivatives of neural activities may code for error signals.

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12. Calculating the Mutual Information between Two Spike Trains

Conor Houghton

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It is difficult to estimate the mutual information between spike trains because established methods require more data than are usually available. Kozachenko-Leonenko estimators promise to solve this problem but include a smoothing parameter that must be set. We propose here that the smoothing parameter can be selected by maximizing the estimated unbiased mutual information. This is tested on fictive data and shown to work very well.

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13. Learning Invariant Features in Modulatory Networks through Conflict and Ambiguity

W. Shane Grant, Laurent Itti

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This work lays the foundation for a framework of cortical learning based on the idea of a competitive column, which is inspired by the functional organization of neurons in the cortex. A column describes a prototypical organization for neurons that gives rise to an ability to learn scale, rotation, and translation-invariant features. This is empowered by a recently developed learning rule, conflict learning, which enables the network to learn over both driving and modulatory feedforward, feedback, and lateral inputs. The framework is further supported by introducing both a notion of neural ambiguity and an adaptive threshold scheme. Ambiguity, which captures the idea that too many decisions lead to indecision, gives the network a dynamic way to resolve locally ambiguous decisions. The adaptive threshold operates over multiple timescales to regulate neural activity under the varied arrival timings of input in a highly interconnected multilayer network with feedforward and feedback. The competitive column architecture is demonstrated on a large-scale (54,000 neurons and 18 million synapses), invariant model of border ownership. The model is trained on four simple, fixed-scale shapes: two squares, one rectangle, and one symmetric L-shape. Tested on 1899 synthetic shapes of varying scale and complexity, the model correctly assigned border ownership with 74% accuracy. The model's abilities were also illustrated on contours of objects taken from natural images. Combined with conflict learning, the competitive column and ambiguity give a better intuitive understanding of how feedback, modulation, and inhibition may interact in the brain to influence activation and learning.

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14. Dynamic Computational Model of the Human Spinal Cord Connectome

Jeffrey E. Arle, N. Iftimia, Jay L. Shils, Longzhi Mei, Kristen W. Carlson

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Connectomes abound, but few for the human spinal cord. Using anatomical data in the literature, we constructed a draft connectivity map of the human spinal cord connectome, providing a template for the many calibrations of specialized behavior to be overlaid on it and the basis for an initial computational model. A thorough literature review gleaned cell types, connectivity, and connection strength indications. Where human data were not available, we selected species that have been studied. Cadaveric spinal cord measurements, cross-sectional histology images, and cytoarchitectural data regarding cell size and density served as the starting point for estimating numbers of neurons. Simulations were run using neural circuitry simulation software. The model contains the neural circuitry in all ten Rexed laminae with intralaminar,

interlaminar, and intersegmental connections, as well as ascending and descending brain connections and estimated neuron counts for various cell types in every lamina of all 31 segments. We noted the presence of highly interconnected complex networks exhibiting several orders of recurrence. The model was used to perform a detailed study of spinal cord stimulation for analgesia. This model is a starting point for workers to develop and test hypotheses across an array of biomedical applications focused on the spinal cord. Each such model requires additional calibrations to constrain its output to verifiable predictions. Future work will include simulating additional segments and expanding the research uses of the model.

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15. Accelerating Nonnegative Matrix Factorization Algorithms Using Extrapolation

Andersen Ang, Nicolas Gillis

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We propose a general framework to accelerate significantly the algorithms for nonnegative matrix factorization (NMF). This framework is inspired from the extrapolation scheme used to accelerate gradient methods in convex optimization and from the method of parallel tangents. However, the use of extrapolation in the context of the exact coordinate descent algorithms tackling the nonconvex NMF problems is novel. We illustrate the performance of this approach on two state-of-the-art NMF algorithms: accelerated hierarchical alternating least squares and alternating nonnegative least squares, using synthetic, image, and document data sets.

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16. Systems of Bounded Rational Agents with Information-Theoretic Constraints

Sebastian Gottwald, Daniel A. Braun

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Specialization and hierarchical organization are important features of efficient collaboration in economical, artificial, and biological systems. Here, we investigate the hypothesis that both features can be explained by the fact that each entity of such a system is limited in a certain way. We propose an information-theoretic approach based on a free energy principle in order to computationally analyze systems of bounded rational agents that deal with such limitations optimally. We find that specialization allows a focus on fewer tasks, thus leading to a more efficient execution, but in turn, it requires coordination in hierarchical structures of specialized experts and coordinating units. Our results suggest that hierarchical architectures of specialized units at lower levels that are coordinated by units at higher levels are optimal, given that each unit's information-processing capability is limited and conforms to constraints on complexity costs.

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17. The Information Bottleneck and Geometric Clustering

DJ Strouse, David J. Schwab

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The information bottleneck (IB) approach to clustering takes a joint distribution [Formula: see text] and maps the data [Formula: see text] to cluster labels [Formula: see text], which retain maximal information about

[Formula: see text] (Tishby, Pereira, & Bialek, 1999). This objective results in an algorithm that clusters data points based on the similarity of their conditional distributions [Formula: see text]. This is in contrast to classic geometric clustering algorithms such as [Formula: see text]-means and gaussian mixture models (GMMs), which take a set of observed data points [Formula: see text] and cluster them based on their geometric (typically Euclidean) distance from one another. Here, we show how to use the deterministic information bottleneck (DIB) (Strouse & Schwab, 2017), a variant of IB, to perform geometric clustering by choosing cluster labels that preserve information about data point location on a smoothed data set. We also introduce a novel intuitive method to choose the number of clusters via kinks in the information curve. We apply this approach to a variety of simple clustering problems, showing that DIB with our model selection procedure recovers the generative cluster labels. We also show that, in particular limits of our model parameters, clustering with DIB and IB is equivalent to [Formula: see text]-means and EM fitting of a GMM with hard and soft assignments, respectively. Thus, clustering with (D)IB generalizes and provides an information-theoretic perspective on these classic algorithms.

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18. Imaging the Spatiotemporal Dynamics of Cognitive Processes at High Temporal Resolution

Idan Tal, Micha Abeles

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This letter presents a noninvasive imaging technique that captures the exact timing and locations of cortical activity sequences that are specific to a cognitive process. These precise spatiotemporal sequences can be detected in the human brain as specific time-position pattern associated with a cognitive task. They are consistent with direct measurements of population activity recorded in nonhuman primates, thus suggesting that specific time-position patterns associated with a cognitive task can be identified. This imaging technique is based on estimating the amplitude of cortical current dipoles from MEG recordings. Although the spatial resolution of these estimations is poor (approximately 2 cm), the temporal resolution is high (milliseconds). We show that within these cortical current dipoles, time points of cortical activation can be identified as brief amplitude undulations and that sequences of these transients repeat with millisecond accuracy, hence making it possible to treat the timing of these transients as point processes. We illustrate the feasibility of finding spatiotemporal templates specific to the cognitive processes associated with following the rhythm of drumbeats that involve the activation at multiple cortical and cerebellar loci. These templates evolve at an accuracy of a few milliseconds. This approach can thus pave the way for new perspectives on the relationships between brain dynamics and cognition.

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19. Identification of Linear and Nonlinear Sensory Processing Circuits from Spiking Neuron Data

Dorian Florescu, Daniel Coca

https://doi.org/10.1162/neco_a_01051 | pp. 670-707

Inferring mathematical models of sensory processing systems directly from input-output observations, while making the fewest assumptions about the model equations and the types of measurements available, is still a major issue in computational neuroscience. This letter introduces two new approaches for identifying sensory circuit models consisting of linear and nonlinear filters in series with spiking neuron models, based

only on the sampled analog input to the filter and the recorded spike train output of the spiking neuron. For an ideal integrate-and-fire neuron model, the first algorithm can identify the spiking neuron parameters as well as the structure and parameters of an arbitrary nonlinear filter connected to it. The second algorithm can identify the parameters of the more general leaky integrate-and-fire spiking neuron model, as well as the parameters of an arbitrary linear filter connected to it. Numerical studies involving simulated and real experimental recordings are used to demonstrate the applicability and evaluate the performance of the proposed algorithms.

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20. Indistinguishable Synapses Lead to Sparse Networks

Joseph Snider

https://doi.org/10.1162/neco_a_01052 | pp. 708-722

Neurons integrate information from many neighbors when they process information. Inputs to a given neuron are thus indistinguishable from one another. Under the assumption that neurons maximize their information storage, indistinguishability is shown to place a strong constraint on the distribution of strengths between neurons. The distribution of individual synapse strengths is found to follow a modified Boltzmann distribution with strength proportional to [Formula: see text]. The model is shown to be consistent with experimental data from Caenorhabditis elegans connectivity and in vivo synaptic strength measurements. The [Formula: see text] dependence helps account for the observation of many zero or weak connections between neurons or sparsity of the neural network.

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21. Cross-Domain Metric and Multiple Kernel Learning Based on Information Theory

Wei Wang, Hao Wang, Chen Zhang, Yang Gao

https://doi.org/10.1162/neco_a_01053 | pp. 820-855

Learning an appropriate distance metric plays a substantial role in the success of many learning machines. Conventional metric learning algorithms have limited utility when the training and test samples are drawn from related but different domains (i.e., source domain and target domain). In this letter, we propose two novel metric learning algorithms for domain adaptation in an information-theoretic setting, allowing for discriminating power transfer and standard learning machine propagation across two domains. In the first one, a cross-domain Mahalanobis distance is learned by combining three goals: reducing the distribution difference between different domains, preserving the geometry of target domain data, and aligning the geometry of source domain data with label information. Furthermore, we devote our efforts to solving complex domain adaptation problems and go beyond linear cross-domain metric learning by extending the first method to a multiple kernel learning framework. A convex combination of multiple kernels and a linear transformation are adaptively learned in a single optimization, which greatly benefits the exploration of prior knowledge and the description of data characteristics. Comprehensive experiments in three real-world applications (face recognition, text classification, and object categorization) verify that the proposed methods outperform state-of-the-art metric learning and domain adaptation methods.

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22. Dynamic Neural Turing Machine with Continuous and Discrete Addressing Schemes

Ça■lar Gülçehre, Sarath Chandar, Kyunghyun Cho, Yoshua Bengio

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We extend the neural Turing machine (NTM) model into a dynamic neural Turing machine (D-NTM) by introducing trainable address vectors. This addressing scheme maintains for each memory cell two separate vectors, content and address vectors. This allows the D-NTM to learn a wide variety of location-based addressing strategies, including both linear and nonlinear ones. We implement the D-NTM with both continuous and discrete read and write mechanisms. We investigate the mechanisms and effects of learning to read and write into a memory through experiments on Facebook bAbI tasks using both a feedforward and GRU controller. We provide extensive analysis of our model and compare different variations of neural Turing machines on this task. We show that our model outperforms long short-term memory and NTM variants. We provide further experimental results on the sequential [Formula: see text]MNIST, Stanford Natural Language Inference, associative recall, and copy tasks.

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23. Information-Theoretic Bounds and Approximations in Neural Population Coding

Wentao Huang, Kechen Zhang

https://doi.org/10.1162/neco_a_01056 | pp. 885-944

While Shannon's mutual information has widespread applications in many disciplines, for practical applications it is often difficult to calculate its value accurately for high-dimensional variables because of the curse of dimensionality. This article focuses on effective approximation methods for evaluating mutual information in the context of neural population coding. For large but finite neural populations, we derive several information-theoretic asymptotic bounds and approximation formulas that remain valid in high-dimensional spaces. We prove that optimizing the population density distribution based on these approximation formulas is a convex optimization problem that allows efficient numerical solutions. Numerical simulation results confirmed that our asymptotic formulas were highly accurate for approximating mutual information for large neural populations. In special cases, the approximation formulas are exactly equal to the true mutual information. We also discuss techniques of variable transformation and dimensionality reduction to facilitate computation of the approximations.

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24. A Unifying Framework of Synaptic and Intrinsic Plasticity in Neural Populations

Johannes Leugering, Gordon Pipa

https://doi.org/10.1162/neco_a_01057 | pp. 945-986

A neuronal population is a computational unit that receives a multivariate, time-varying input signal and creates a related multivariate output. These neural signals are modeled as stochastic processes that transmit information in real time, subject to stochastic noise. In a stationary environment, where the input signals can be characterized by constant statistical properties, the systematic relationship between its input and output processes determines the computation carried out by a population. When these statistical

characteristics unexpectedly change, the population needs to adapt to its new environment if it is to maintain stable operation. Based on the general concept of homeostatic plasticity, we propose a simple compositional model of adaptive networks that achieve invariance with regard to undesired changes in the statistical properties of their input signals and maintain outputs with well-defined joint statistics. To achieve such invariance, the network model combines two functionally distinct types of plasticity. An abstract stochastic process neuron model implements a generalized form of intrinsic plasticity that adapts marginal statistics, relying only on mechanisms locally confined within each neuron and operating continuously in time, while a simple form of Hebbian synaptic plasticity operates on synaptic connections, thus shaping the interrelation between neurons as captured by a copula function. The combined effect of both mechanisms allows a neuron population to discover invariant representations of its inputs that remain stable under a wide range of transformations (e.g., shifting, scaling and (affine linear) mixing). The probabilistic model of homeostatic adaptation on a population level as presented here allows us to isolate and study the individual and the interaction dynamics of both mechanisms of plasticity and could guide the future search for computationally beneficial types of adaptation.

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25. Robust Modulation of Integrate-and-Fire Models

Tomas Van Pottelbergh, Guillaume Drion, Rodolphe Sepulchre
https://doi.org/10.1162/neco_a_01065 | pp. 987-1011

By controlling the state of neuronal populations, neuromodulators ultimately affect behavior. A key neuromodulation mechanism is the alteration of neuronal excitability via the modulation of ion channel expression. This type of neuromodulation is normally studied with conductance-based models, but those models are computationally challenging for large-scale network simulations needed in population studies. This article studies the modulation properties of the multiquadratic integrate-and-fire model, a generalization of the classical quadratic integrate-and-fire model. The model is shown to combine the computational economy of integrate-and-fire modeling and the physiological interpretability of conductance-based modeling. It is therefore a good candidate for affordable computational studies of neuromodulation in large networks.

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26. Dethroning the Fano Factor: A Flexible, Model-Based Approach to Partitioning Neural Variability

Adam S. Charles, Mijung Park, J. Patrick Weller, Gregory D. Horwitz, Jonathan W. Pillow
https://doi.org/10.1162/neco_a_01062 | pp. 1012-1045

Neurons in many brain areas exhibit high trial-to-trial variability, with spike counts that are overdispersed relative to a Poisson distribution. Recent work (Goris, Movshon, & Simoncelli, 2014) has proposed to explain this variability in terms of a multiplicative interaction between a stochastic gain variable and a stimulus-dependent Poisson firing rate, which produces quadratic relationships between spike count mean and variance. Here we examine this quadratic assumption and propose a more flexible family of models that can account for a more diverse set of mean-variance relationships. Our model contains additive gaussian noise that is transformed nonlinearly to produce a Poisson spike rate. Different choices of the nonlinear function can give rise to qualitatively different mean-variance relationships, ranging from sublinear to linear to quadratic. Intriguingly, a rectified squaring nonlinearity produces a linear mean-variance function,

corresponding to responses with a constant Fano factor. We describe a computationally efficient method for fitting this model to data and demonstrate that a majority of neurons in a V1 population are better described by a model with a nonquadratic relationship between mean and variance. Finally, we demonstrate a practical use of our model via an application to Bayesian adaptive stimulus selection in closed-loop neurophysiology experiments, which shows that accounting for overdispersion can lead to dramatic improvements in adaptive tuning curve estimation.

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27. Estimating a Separably Markov Random Field from Binary Observations

Yingzhuo Zhang, Noa Malem-Shinitski, Stephen A. Allsop, Kay M. Tye, Demba Ba
https://doi.org/10.1162/neco_a_01059 | pp. 1046-1079

A fundamental problem in neuroscience is to characterize the dynamics of spiking from the neurons in a circuit that is involved in learning about a stimulus or a contingency. A key limitation of current methods to analyze neural spiking data is the need to collapse neural activity over time or trials, which may cause the loss of information pertinent to understanding the function of a neuron or circuit. We introduce a new method that can determine not only the trial-to-trial dynamics that accompany the learning of a contingency by a neuron, but also the latency of this learning with respect to the onset of a conditioned stimulus. The backbone of the method is a separable two-dimensional (2D) random field (RF) model of neural spike rasters, in which the joint conditional intensity function of a neuron over time and trials depends on two latent Markovian state sequences that evolve separately but in parallel. Classical tools to estimate state-space models cannot be applied readily to our 2D separable RF model. We develop efficient statistical and computational tools to estimate the parameters of the separable 2D RF model. We apply these to data collected from neurons in the prefrontal cortex in an experiment designed to characterize the neural underpinnings of the associative learning of fear in mice. Overall, the separable 2D RF model provides a detailed, interpretable characterization of the dynamics of neural spiking that accompany the learning of a contingency.

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28. Controlling Complexity of Cerebral Cortex Simulations—I: CxSystem, a Flexible Cortical Simulation Framework

Vafa Andalibi, Henri Hokkanen, Simo Vanni
https://doi.org/10.1162/neco_a_01120 | pp. 1048-1065

Simulation of the cerebral cortex requires a combination of extensive domain-specific knowledge and efficient software. However, when the complexity of the biological system is combined with that of the software, the likelihood of coding errors increases, which slows model adjustments. Moreover, few life scientists are familiar with software engineering and would benefit from simplicity in form of a high-level abstraction of the biological model. Our primary aim was to build a scalable cortical simulation framework for personal computers. We isolated an adjustable part of the domain-specific knowledge from the software. Next, we designed a framework that reads the model parameters from comma-separated value files and creates the necessary code for Brian2 model simulation. This separation allows rapid exploration of complex cortical circuits while decreasing the likelihood of coding errors and automatically using efficient hardware devices. Next, we tested the system on a simplified version of the neocortical microcircuit proposed by Markram and colleagues (2015). Our results indicate that the framework can efficiently perform simulations

using Python, C[Formula: see text], and GPU devices. The most efficient device varied with computer hardware and the duration and scale of the simulated system. The speed of Brian2 was retained despite an overlying layer of software. However, the Python and C[Formula: see text] devices inherited the single core limitation of Brian2. The CxSystem framework supports exploration of complex models on personal computers and thus has the potential to facilitate research on cortical networks and systems.

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29. Adaptive Structure Concept Factorization for Multiview Clustering

Kun Zhan, Jinhui Shi, Jing Wang, Haibo Wang, Yuange Xie

https://doi.org/10.1162/neco_a_01055 | pp. 1080-1103

Most existing multiview clustering methods require that graph matrices in different views are computed beforehand and that each graph is obtained independently. However, this requirement ignores the correlation between multiple views. In this letter, we tackle the problem of multiview clustering by jointly optimizing the graph matrix to make full use of the data correlation between views. With the interview correlation, a concept factorization-based multiview clustering method is developed for data integration, and the adaptive method correlates the affinity weights of all views. This method differs from nonnegative matrix factorization-based clustering methods in that it can be applicable to data sets containing negative values. Experiments are conducted to demonstrate the effectiveness of the proposed method in comparison with state-of-the-art approaches in terms of accuracy, normalized mutual information, and purity.

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30. Muscle Synergy–Driven Robust Motion Control

Kyuengbo Min, Masami Iwamoto, Shinji Kakei, Hideyuki Kimpara

https://doi.org/10.1162/neco_a_01063 | pp. 1104-1131

Humans are able to robustly maintain desired motion and posture under dynamically changing circumstances, including novel conditions. To accomplish this, the brain needs to optimize the synergistic control between muscles against external dynamic factors. However, previous related studies have usually simplified the control of multiple muscles using two opposing muscles, which are minimum actuators to simulate linear feedback control. As a result, they have been unable to analyze how muscle synergy contributes to motion control robustness in a biological system. To address this issue, we considered a new muscle synergy concept used to optimize the synergy between muscle units against external dynamic conditions, including novel conditions. We propose that two main muscle control policies synergistically control muscle units to maintain the desired motion against external dynamic conditions. Our assumption is based on biological evidence regarding the control of multiple muscles via the corticospinal tract. One of the policies is the group control policy (GCP), which is used to control muscle group units classified based on functional similarities in joint control. This policy is used to effectively resist external dynamic circumstances, such as disturbances. The individual control policy (ICP) assists the GCP in precisely controlling motion by controlling individual muscle units. To validate this hypothesis, we simulated the reinforcement of the synergistic actions of the two control policies during the reinforcement learning of feedback motion control. Using this learning paradigm, the two control policies were synergistically combined to result in robust feedback control under novel transient and sustained disturbances that did not involve learning. Further, by comparing our data to experimental data generated by human subjects under the same conditions as those of the simulation, we showed that the proposed synergy concept may be used to analyze muscle

synergy-driven motion control robustness in humans.

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31. Coexistence of Stochastic Oscillations and Self-Organized Criticality in a Neuronal Network: Sandpile Model Application

Alireza Saeedi, Mostafa Jannesari, Shahriar Gharibzadeh, Fatemeh Bakouie

https://doi.org/10.1162/neco_a_01061 | pp. 1132-1149

Self-organized criticality (SOC) and stochastic oscillations (SOs) are two theoretically contradictory phenomena that are suggested to coexist in the brain. Recently it has been shown that an accumulation-release process like sandpile dynamics can generate SOC and SOs simultaneously. We considered the effect of the network structure on this coexistence and showed that the sandpile dynamics on a small-world network can produce two power law regimes along with two groups of SOs-two peaks in the power spectrum of the generated signal simultaneously. We also showed that external stimuli in the sandpile dynamics do not affect the coexistence of SOC and SOs but increase the frequency of SOs, which is consistent with our knowledge of the brain.

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32. Slowness as a Proxy for Temporal Predictability: An Empirical Comparison

Björn Weghenkel, Laurenz Wiskott

https://doi.org/10.1162/neco_a_01070 | pp. 1151-1179

The computational principles of slowness and predictability have been proposed to describe aspects of information processing in the visual system. From the perspective of slowness being a limited special case of predictability we investigate the relationship between these two principles empirically. On a collection of real-world data sets we compare the features extracted by slow feature analysis (SFA) to the features of three recently proposed methods for predictable feature extraction: forecastable component analysis, predictable feature analysis, and graph-based predictable feature analysis. Our experiments show that the predictability of the learned features is highly correlated, and, thus, SFA appears to effectively implement a method for extracting predictable features according to different measures of predictability.

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33. Designing Patient-Specific Optimal Neurostimulation Patterns for Seizure Suppression

Roman Sandler, Kunling Geng, Dong Song, Robert E. Hampson, Mark R. Witcher, Sam A. Deadwyler, Theodore W. Berger, Vasilis Z. Marmarelis

https://doi.org/10.1162/neco_a_01075 | pp. 1180-1208

Neurostimulation is a promising therapy for abating epileptic seizures. However, it is extremely difficult to identify optimal stimulation patterns experimentally. In this study, human recordings are used to develop a functional 24 neuron network statistical model of hippocampal connectivity and dynamics. Spontaneous seizure-like activity is induced in silico in this reconstructed neuronal network. The network is then used as a testbed to design and validate a wide range of neurostimulation patterns. Commonly used periodic trains were not able to permanently abate seizures at any frequency. A simulated annealing global optimization

algorithm was then used to identify an optimal stimulation pattern, which successfully abated 92% of seizures. Finally, in a fully responsive, or closed-loop, neurostimulation paradigm, the optimal stimulation successfully prevented the network from entering the seizure state. We propose that the framework presented here for algorithmically identifying patient-specific neurostimulation patterns can greatly increase the efficacy of neurostimulation devices for seizures.

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34. Predictive Coding in Area V4: Dynamic Shape Discrimination under Partial Occlusion

Hannah Choi, Anitha Pasupathy, Eric Shea■Brown

https://doi.org/10.1162/neco_a_01072 | pp. 1209-1257

The primate visual system has an exquisite ability to discriminate partially occluded shapes. Recent electrophysiological recordings suggest that response dynamics in intermediate visual cortical area V4, shaped by feedback from prefrontal cortex (PFC), may play a key role. To probe the algorithms that may underlie these findings, we build and test a model of V4 and PFC interactions based on a hierarchical predictive coding framework. We propose that probabilistic inference occurs in two steps. Initially, V4 responses are driven solely by bottom-up sensory input and are thus strongly influenced by the level of occlusion. After a delay, V4 responses combine both feedforward input and feedback signals from the PFC; the latter reflect predictions made by PFC about the visual stimulus underlying V4 activity. We find that this model captures key features of V4 and PFC dynamics observed in experiments. Specifically, PFC responses are strongest for occluded stimuli and delayed responses in V4 are less sensitive to occlusion, supporting our hypothesis that the feedback signals from PFC underlie robust discrimination of occluded shapes. Thus, our study proposes that area V4 and PFC participate in hierarchical inference, with feedback signals encoding top-down predictions about occluded shapes.

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35. Pattern Storage, Bifurcations, and Groupwise Correlation Structure of an Exactly Solvable Asymmetric Neural Network Model

Diego Fasoli, Anna Cattani, Stefano Panzeri

https://doi.org/10.1162/neco_a_01069 | pp. 1258-1295

Despite their biological plausibility, neural network models with asymmetric weights are rarely solved analytically, and closed-form solutions are available only in some limiting cases or in some mean-field approximations. We found exact analytical solutions of an asymmetric spin model of neural networks with arbitrary size without resorting to any approximation, and we comprehensively studied its dynamical and statistical properties. The network had discrete time evolution equations and binary firing rates, and it could be driven by noise with any distribution. We found analytical expressions of the conditional and stationary joint probability distributions of the membrane potentials and the firing rates. By manipulating the conditional probability distribution of the firing rates, we extend to stochastic networks the associating learning rule previously introduced by Personnaz and coworkers. The new learning rule allowed the safe storage, under the presence of noise, of point and cyclic attractors, with useful implications for content-addressable memories. Furthermore, we studied the bifurcation structure of the network dynamics in the zero-noise limit. We analytically derived examples of the codimension 1 and codimension 2 bifurcation diagrams of the network, which describe how the neuronal dynamics changes with the external stimuli. This showed that the

network may undergo transitions among multistable regimes, oscillatory behavior elicited by asymmetric synaptic connections, and various forms of spontaneous symmetry breaking. We also calculated analytically groupwise correlations of neural activity in the network in the stationary regime. This revealed neuronal regimes where, statistically, the membrane potentials and the firing rates are either synchronous or asynchronous. Our results are valid for networks with any number of neurons, although our equations can be realistically solved only for small networks. For completeness, we also derived the network equations in the thermodynamic limit of infinite network size and we analytically studied their local bifurcations. All the analytical results were extensively validated by numerical simulations.

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36. An Interneuron Circuit Reproducing Essential Spectral Features of Field Potentials

Reinoud Maex

https://doi.org/10.1162/neco_a_01068 | pp. 1296-1322

Recent advances in engineering and signal processing have renewed the interest in invasive and surface brain recordings, yet many features of cortical field potentials remain incompletely understood. In the computational study that follows, we show that a model circuit of interneurons, coupled via both GABA A receptor synapses and electrical synapses, reproduces many essential features of the power spectrum of local field potential (LFP) recordings, such as 1/ f power scaling at low frequency (below 10 Hz), power accumulation in the γ-frequency band (30–100 Hz), and a robust α rhythm in the absence of stimulation. The low-frequency 1/ f power scaling depends on strong reciprocal inhibition, whereas the α rhythm is generated by electrical coupling of intrinsically active neurons. As in previous studies, the γ power arises through the amplification of single-neuron spectral properties, owing to the refractory period, by parameters that favor neuronal synchrony, such as delayed inhibition. This study also confirms that both synaptic and voltage-gated membrane currents contribute substantially to the LFP and that high-frequency signals such as action potentials quickly taper off with distance. Given the ubiquity of electrically coupled interneuron circuits in the mammalian brain, they may be major determinants of the recorded potentials.

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37. Optimizing the Usability of Brain-Computer Interfaces

Yin Zhang■, Steven M. Chase

https://doi.org/10.1162/neco_a_01076 | pp. 1323-1358

Brain-computer interfaces are in the process of moving from the laboratory to the clinic. These devices act by reading neural activity and using it to directly control a device, such as a cursor on a computer screen. An open question in the field is how to map neural activity to device movement in order to achieve the most proficient control. This question is complicated by the fact that learning, especially the long-term skill learning that accompanies weeks of practice, can allow subjects to improve performance over time. Typical approaches to this problem attempt to maximize the biomimetic properties of the device in order to limit the need for extensive training. However, it is unclear if this approach would ultimately be superior to performance that might be achieved with a nonbiomimetic device once the subject has engaged in extended practice and learned how to use it. Here we approach this problem using ideas from optimal control theory. Under the assumption that the brain acts as an optimal controller, we present a formal definition of the usability of a device and show that the optimal postlearning mapping can be written as the solution of a

constrained optimization problem. We then derive the optimal mappings for particular cases common to most brain-computer interfaces. Our results suggest that the common approach of creating biomimetic interfaces may not be optimal when learning is taken into account. More broadly, our method provides a blueprint for optimal device design in general control-theoretic contexts.

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38. Solving Constraint-Satisfaction Problems with Distributed Neocortical-Like Neuronal Networks

Ueli Rutishauser, Jean-Jacques Slotine, Rodney J. Douglas

https://doi.org/10.1162/neco_a_01074 | pp. 1359-1393

Finding actions that satisfy the constraints imposed by both external inputs and internal representations is central to decision making. We demonstrate that some important classes of constraint satisfaction problems (CSPs) can be solved by networks composed of homogeneous cooperative-competitive modules that have connectivity similar to motifs observed in the superficial layers of neocortex. The winner-take-all modules are sparsely coupled by programming neurons that embed the constraints onto the otherwise homogeneous modular computational substrate. We show rules that embed any instance of the CSP's planar four-color graph coloring, maximum independent set, and sudoku on this substrate and provide mathematical proofs that guarantee these graph coloring problems will convergence to a solution. The network is composed of nonsaturating linear threshold neurons. Their lack of right saturation allows the overall network to explore the problem space driven through the unstable dynamics generated by recurrent excitation. The direction of exploration is steered by the constraint neurons. While many problems can be solved using only linear inhibitory constraints, network performance on hard problems benefits significantly when these negative constraints are implemented by nonlinear multiplicative inhibition. Overall, our results demonstrate the importance of instability rather than stability in network computation and offer insight into the computational role of dual inhibitory mechanisms in neural circuits.

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39. A Kernel Embedding–Based Approach for Nonstationary Causal Model Inference

Shoubo Hu, Zhitang Chen, Laiwan Chan

https://doi.org/10.1162/neco_a_01064 | pp. 1394-1425

Although nonstationary data are more common in the real world, most existing causal discovery methods do not take nonstationarity into consideration. In this letter, we propose a kernel embedding-based approach, ENCI, for nonstationary causal model inference where data are collected from multiple domains with varying distributions. In ENCI, we transform the complicated relation of a cause-effect pair into a linear model of variables of which observations correspond to the kernel embeddings of the cause-and-effect distributions in different domains. In this way, we are able to estimate the causal direction by exploiting the causal asymmetry of the transformed linear model. Furthermore, we extend ENCI to causal graph discovery for multiple variables by transforming the relations among them into a linear nongaussian acyclic model. We show that by exploiting the nonstationarity of distributions, both cause-effect pairs and two kinds of causal graphs are identifiable under mild conditions. Experiments on synthetic and real-world data are conducted to justify the efficacy of ENCI over major existing methods.

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40. Deep Semisupervised Zero-Shot Learning with Maximum Mean Discrepancy

Lingling Zhang, Jun Liu, Minnan Luo, Xiaojun Chang, Qinghua Zheng

https://doi.org/10.1162/neco_a_01071 | pp. 1426-1447

Due to the difficulty of collecting labeled images for hundreds of thousands of visual categories, zero-shot learning, where unseen categories do not have any labeled images in training stage, has attracted more attention. In the past, many studies focused on transferring knowledge from seen to unseen categories by projecting all category labels into a semantic space. However, the label embeddings could not adequately express the semantics of categories. Furthermore, the common semantics of seen and unseen instances cannot be captured accurately because the distribution of these instances may be quite different. For these issues, we propose a novel deep semisupervised method by jointly considering the heterogeneity gap between different modalities and the correlation among unimodal instances. This method replaces the original labels with the corresponding textual descriptions to better capture the category semantics. This method also overcomes the problem of distribution difference by minimizing the maximum mean discrepancy between seen and unseen instance distributions. Extensive experimental results on two benchmark data sets, CU200-Birds and Oxford Flowers-102, indicate that our method achieves significant improvements over previous methods.

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41. A Theory of Sequence Indexing and Working Memory in Recurrent Neural Networks

E. Paxon Frady, Denis Kleyko, Friedrich T. Sommer

https://doi.org/10.1162/neco_a_01084 | pp. 1449-1513

To accommodate structured approaches of neural computation, we propose a class of recurrent neural networks for indexing and storing sequences of symbols or analog data vectors. These networks with randomized input weights and orthogonal recurrent weights implement coding principles previously described in vector symbolic architectures (VSA) and leverage properties of reservoir computing. In general, the storage in reservoir computing is lossy, and crosstalk noise limits the retrieval accuracy and information capacity. A novel theory to optimize memory performance in such networks is presented and compared with simulation experiments. The theory describes linear readout of analog data and readout with winner-take-all error correction of symbolic data as proposed in VSA models. We find that diverse VSA models from the literature have universal performance properties, which are superior to what previous analyses predicted. Further, we propose novel VSA models with the statistically optimal Wiener filter in the readout that exhibit much higher information capacity, in particular for storing analog data. The theory we present also applies to memory buffers, networks with gradual forgetting, which can operate on infinite data streams without memory overflow. Interestingly, we find that different forgetting mechanisms, such as attenuating recurrent weights or neural nonlinearities, produce very similar behavior if the forgetting time constants are matched. Such models exhibit extensive capacity when their forgetting time constant is optimized for given noise conditions and network size. These results enable the design of new types of VSA models for the online processing of data streams.

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42. SuperSpike: Supervised Learning in Multilayer Spiking Neural Networks

Friedemann Zenke, Surya Ganguli
https://doi.org/10.1162/neco_a_01086 | pp. 1514-1541

A vast majority of computation in the brain is performed by spiking neural networks. Despite the ubiquity of such spiking, we currently lack an understanding of how biological spiking neural circuits learn and compute in vivo, as well as how we can instantiate such capabilities in artificial spiking circuits in silico. Here we revisit the problem of supervised learning in temporally coding multilayer spiking neural networks. First, by using a surrogate gradient approach, we derive SuperSpike, a nonlinear voltage-based three-factor learning rule capable of training multilayer networks of deterministic integrate-and-fire neurons to perform nonlinear computations on spatiotemporal spike patterns. Second, inspired by recent results on feedback alignment, we compare the performance of our learning rule under different credit assignment strategies for propagating output errors to hidden units. Specifically, we test uniform, symmetric, and random feedback, finding that simpler tasks can be solved with any type of feedback, while more complex tasks require symmetric feedback. In summary, our results open the door to obtaining a better scientific understanding of learning and computation in spiking neural networks by advancing our ability to train them to solve nonlinear problems involving transformations between different spatiotemporal spike time patterns.

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43. A Learning Framework for Winner-Take-All Networks with Stochastic Synapses

Hesham Mostafa, Gert Cauwenberghs
https://doi.org/10.1162/neco_a_01080 | pp. 1542-1572

Many recent generative models make use of neural networks to transform the probability distribution of a simple low-dimensional noise process into the complex distribution of the data. This raises the question of whether biological networks operate along similar principles to implement a probabilistic model of the environment through transformations of intrinsic noise processes. The intrinsic neural and synaptic noise processes in biological networks, however, are quite different from the noise processes used in current abstract generative networks. This, together with the discrete nature of spikes and local circuit interactions among the neurons, raises several difficulties when using recent generative modeling frameworks to train biologically motivated models. In this letter, we show that a biologically motivated model based on multilayer winner-take-all circuits and stochastic synapses admits an approximate analytical description. This allows us to use the proposed networks in a variational learning setting where stochastic backpropagation is used to optimize a lower bound on the data log likelihood, thereby learning a generative model of the data. We illustrate the generality of the proposed networks and learning technique by using them in a structured output prediction task and a semisupervised learning task. Our results extend the domain of application of modern stochastic network architectures to networks where synaptic transmission failure is the principal noise mechanism.

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44. Optimal Readout of Correlated Neural Activity in a Decision-Making Circuit

Matias Calderini, Sophie Zhang, Nareg Berberian, Jean■Philippe Thivierge
https://doi.org/10.1162/neco_a_01083 | pp. 1573-1611

The neural correlates of decision making have been extensively studied with tasks involving a choice between two alternatives that is guided by visual cues. While a large body of work argues for a role of the lateral intraparietal (LIP) region of cortex in these tasks, this role may be confounded by the interaction between LIP and other regions, including medial temporal (MT) cortex. Here, we describe a simplified linear model of decision making that is adapted to two tasks: a motion discrimination and a categorization task. We show that the distinct contribution of MT and LIP may indeed be confounded in these tasks. In particular, we argue that the motion discrimination task relies on a straightforward visuomotor mapping, which leads to redundant information between MT and LIP. The categorization task requires a more complex mapping between visual information and decision behavior, and therefore does not lead to redundancy between MT and LIP. Going further, the model predicts that noise correlations within LIP should be greater in the categorization compared to the motion discrimination task due to the presence of shared inputs from MT. The impact of these correlations on task performance is examined by analytically deriving error estimates of an optimal linear readout for shared and unique inputs. Taken together, results clarify the contribution of MT and LIP to decision making and help characterize the role of noise correlations in these regions.

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45. Novel Perceptually Uniform Chromatic Space

María da Fonseca, Inés Samengo

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Chromatically perceptive observers are endowed with a sense of similarity between colors. For example, two shades of green that are only slightly discriminable are perceived as similar, whereas other pairs of colors, for example, blue and yellow, typically elicit markedly different sensations. The notion of similarity need not be shared by different observers. Dichromat and trichromat subjects perceive colors differently, and two dichromats (or two trichromats, for that matter) may judge chromatic differences inconsistently. Moreover, there is ample evidence that different animal species sense colors diversely. To capture the subjective metric of color perception, here we construct a notion of distance in color space based on the physiology of the retina, and is thereby individually tailored for different observers. By applying the Fisher metric to an analytical model of color representation, we construct a notion of distance that reproduces behavioral experiments of classical discrimination tasks. We then derive a coordinate transformation that defines a new chromatic space in which the Euclidean distance between any two colors is equal to the perceptual distance, as seen by one individual subject, endowed with an arbitrary number of color-sensitive photoreceptors, each with arbitrary absorption probability curves and appearing in arbitrary proportions.

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46. Robust MST-Based Clustering Algorithm

Qidong Liu, Ruisheng Zhang, Zhili Zhao, Zhenghai Wang, Mengyao Jiao, Guangjing Wang

https://doi.org/10.1162/neco_a_01081 | pp. 1624-1646

Minimax similarity stresses the connectedness of points via mediating elements rather than favoring high mutual similarity. The grouping principle yields superior clustering results when mining arbitrarily-shaped clusters in data. However, it is not robust against noises and outliers in the data. There are two main problems with the grouping principle: first, a single object that is far away from all other objects defines a separate cluster, and second, two connected clusters would be regarded as two parts of one cluster. In order to solve such problems, we propose robust minimum spanning tree (MST)-based clustering algorithm

in this letter. First, we separate the connected objects by applying a density-based coarsening phase, resulting in a low-rank matrix in which the element denotes the supernode by combining a set of nodes. Then a greedy method is presented to partition those supernodes through working on the low-rank matrix. Instead of removing the longest edges from MST, our algorithm groups the data set based on the minimax similarity. Finally, the assignment of all data points can be achieved through their corresponding supernodes. Experimental results on many synthetic and real-world data sets show that our algorithm consistently outperforms compared clustering algorithms.

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47. Facet Annotation by Extending CNN with a Matching Strategy

Bei Wu, Bifan Wei, Jun Liu, Zhaotong Guo, Yuanhao Zheng, Yihe Chen

https://doi.org/10.1162/neco_a_01077 | pp. 1647-1672

Most community question answering (CQA) websites manage plenty of question-answer pairs (QAPs) through topic-based organizations, which may not satisfy users' fine-grained search demands. Facets of topics serve as a powerful tool to navigate, refine, and group the QAPs. In this work, we propose FACM, a model to annotate QAPs with facets by extending convolution neural networks (CNNs) with a matching strategy. First, phrase information is incorporated into text representation by CNNs with different kernel sizes. Then, through a matching strategy among QAPs and facet label texts (FaLTs) acquired from Wikipedia, we generate similarity matrices to deal with the facet heterogeneity. Finally, a three-channel CNN is trained for facet label assignment of QAPs. Experiments on three real-world data sets show that FACM outperforms the state-of-the-art methods.

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48. Distributed Newton Methods for Deep Neural Networks

Chien-Chih Wang, Kent Loong Tan, Chun■Ting Chen, Yu-Hsiang Lin, S. Sathiya Keerthi, Dhruv Mahajan, S. Sundararajan, Chih■Jen Lin

https://doi.org/10.1162/neco_a_01088 | pp. 1673-1724

Deep learning involves a difficult nonconvex optimization problem with a large number of weights between any two adjacent layers of a deep structure. To handle large data sets or complicated networks, distributed training is needed, but the calculation of function, gradient, and Hessian is expensive. In particular, the communication and the synchronization cost may become a bottleneck. In this letter, we focus on situations where the model is distributedly stored and propose a novel distributed Newton method for training deep neural networks. By variable and feature-wise data partitions and some careful designs, we are able to explicitly use the Jacobian matrix for matrix-vector products in the Newton method. Some techniques are incorporated to reduce the running time as well as memory consumption. First, to reduce the communication cost, we propose a diagonalization method such that an approximate Newton direction can be obtained without communication between machines. Second, we consider subsampled Gauss-Newton matrices for reducing the running time as well as the communication cost. Third, to reduce the synchronization cost, we terminate the process of finding an approximate Newton direction even though some nodes have not finished their tasks. Details of some implementation issues in distributed environments are thoroughly investigated. Experiments demonstrate that the proposed method is effective for the distributed training of deep neural networks. Compared with stochastic gradient methods, it is more robust and may give better test accuracy.

49. Joint Estimation of Effective Brain Wave Activation Modes Using EEG/MEG Sensor Arrays and Multimodal MRI Volumes

V. L. Galinsky, Ant■■gona Mart■■nez, Martin P. Paulus, Lawrence R. Frank

https://doi.org/10.1162/neco_a_01087 | pp. 1725-1749

In this letter, we present a new method for integration of sensor-based multifrequency bands of electroencephalography and magnetoencephalography data sets into a voxel-based structural-temporal magnetic resonance imaging analysis by utilizing the general joint estimation using entropy regularization (JESTER) framework. This allows enhancement of the spatial-temporal localization of brain function and the ability to relate it to morphological features and structural connectivity. This method has broad implications for both basic neuroscience research and clinical neuroscience focused on identifying disease-relevant biomarkers by enhancing the spatial-temporal resolution of the estimates derived from current neuroimaging modalities, thereby providing a better picture of the normal human brain in basic neuroimaging experiments and variations associated with disease states.

50. t-SNE Visualization of Large-Scale Neural Recordings

George Dimitriadis, Joana P. Neto, Adam R. Kampff

https://doi.org/10.1162/neco_a_01097 | pp. 1750-1774

Electrophysiology is entering the era of big data. Multiple probes, each with hundreds to thousands of individual electrodes, are now capable of simultaneously recording from many brain regions. The major challenge confronting these new technologies is transforming the raw data into physiologically meaningful signals, that is, single unit spikes. Sorting the spike events of individual neurons from a spatiotemporally dense sampling of the extracellular electric field is a problem that has attracted much attention (Rey, Pedreira, & Quian Quiroga, 2015 ; Rossant et al., 2016) but is still far from solved. Current methods still rely on human input and thus become unfeasible as the size of the data sets grows exponentially. Here we introduce the [Formula: see text]-student stochastic neighbor embedding (t-SNE) dimensionality reduction method (Van der Maaten & Hinton, 2008) as a visualization tool in the spike sorting process. t-SNE embeds the [Formula: see text]-dimensional extracellular spikes ([Formula: see text] = number of features by which each spike is decomposed) into a low- (usually two-) dimensional space. We show that such embeddings, even starting from different feature spaces, form obvious clusters of spikes that can be easily visualized and manually delineated with a high degree of precision. We propose that these clusters represent single units and test this assertion by applying our algorithm on labeled data sets from both hybrid (Rossant et al., 2016) and paired juxtacellular/extracellular recordings (Neto et al., 2016). We have released a graphical user interface (GUI) written in Python as a tool for the manual clustering of the t-SNE embedded spikes and as a tool for an informed overview and fast manual curation of results from different clustering algorithms. Furthermore, the generated visualizations offer evidence in favor of the use of probes with higher density and smaller electrodes. They also graphically demonstrate the diverse nature of the sorting problem when spikes are recorded with different methods and arise from regions with different background spiking statistics.

51. Fully Convolutional Network-Based Multifocus Image Fusion

Xiaopeng Guo, Rencan Nie, Jinde Cao, Dongming Zhou, Wenhua Qian

https://doi.org/10.1162/neco_a_01098 | pp. 1775-1800

As the optical lenses for cameras always have limited depth of field, the captured images with the same scene are not all in focus. Multifocus image fusion is an efficient technology that can synthesize an all-in-focus image using several partially focused images. Previous methods have accomplished the fusion task in spatial or transform domains. However, fusion rules are always a problem in most methods. In this letter, from the aspect of focus region detection, we propose a novel multifocus image fusion method based on a fully convolutional network (FCN) learned from synthesized multifocus images. The primary novelty of this method is that the pixel-wise focus regions are detected through a learning FCN, and the entire image, not just the image patches, are exploited to train the FCN. First, we synthesize 4500 pairs of multifocus images by repeatedly using a gaussian filter for each image from PASCAL VOC 2012, to train the FCN. After that, a pair of source images is fed into the trained FCN, and two score maps indicating the focus property are generated. Next, an inversed score map is averaged with another score map to produce an aggregative score map, which take full advantage of focus probabilities in two score maps. We implement the fully connected conditional random field (CRF) on the aggregative score map to accomplish and refine a binary decision map for the fusion task. Finally, we exploit the weighted strategy based on the refined decision map to produce the fused image. To demonstrate the performance of the proposed method, we compare its fused results with several start-of-the-art methods not only on a gray data set but also on a color data set. Experimental results show that the proposed method can achieve superior fusion performance in both human visual quality and objective assessment.

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52. A Computational Account of the Role of Cochlear Nucleus and Inferior Colliculus in Stabilizing Auditory Nerve Firing for Auditory Category Learning

Irina Higgins, Simon M. Stringer, Jan W. H. Schnupp

https://doi.org/10.1162/neco_a_01085 | pp. 1801-1829

It is well known that auditory nerve (AN) fibers overcome bandwidth limitations through the volley principle, a form of multiplexing. What is less well known is that the volley principle introduces a degree of unpredictability into AN neural firing patterns that may be affecting even simple stimulus categorization learning. We use a physiologically grounded, unsupervised spiking neural network model of the auditory brain with spike time dependent plasticity learning to demonstrate that plastic auditory cortex is unable to learn even simple auditory object categories when exposed to the raw AN firing input without subcortical preprocessing. We then demonstrate the importance of nonplastic subcortical preprocessing within the cochlear nucleus and the inferior colliculus for stabilizing and denoising AN responses. Such preprocessing enables the plastic auditory cortex to learn efficient robust representations of the auditory object categories. The biological realism of our model makes it suitable for generating neurophysiologically testable hypotheses.

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53. Hidden Quantum Processes, Quantum Ion Channels, and 1/f^θ-Type Noise

Alan Paris, Azadeh Vosoughi, Stephen Berman, George Atia
https://doi.org/10.1162/neco_a_01067 | pp. 1830-1929

In this letter, we perform a complete and in-depth analysis of Lorentzian noises, such as those arising from [Formula: see text] and [Formula: see text] channel kinetics, in order to identify the source of [Formula: see text]-type noise in neurological membranes. We prove that the autocovariance of Lorentzian noise depends solely on the eigenvalues (time constants) of the kinetic matrix but that the Lorentzian weighting coefficients depend entirely on the eigenvectors of this matrix. We then show that there are rotations of the kinetic eigenvectors that send any initial weights to any target weights without altering the time constants. In particular, we show there are target weights for which the resulting Lorenztian noise has an approximately [Formula: see text]-type spectrum. We justify these kinetic rotations by introducing a quantum mechanical formulation of membrane stochastics, called hidden quantum activated-measurement models, and prove that these quantum models are probabilistically indistinguishable from the classical hidden Markov models typically used for ion channel stochastics. The quantum dividend obtained by replacing classical with quantum membranes is that rotations of the Lorentzian weights become simple readjustments of the quantum state without any change to the laboratory-determined kinetic and conductance parameters. Moreover, the quantum formalism allows us to model the activation energy of a membrane, and we show that maximizing entropy under constrained activation energy yields the previous [Formula: see text]-type Lorentzian weights, in which the spectral exponent [Formula: see text] is a Lagrange multiplier for the energy constraint. Thus, we provide a plausible neurophysical mechanism by which channel and membrane kinetics can give rise to [Formula: see text]-type noise (something that has been occasionally denied in the literature), as well as a realistic and experimentally testable explanation for the numerical values of the spectral exponents. We also discuss applications of quantum membranes beyond [Formula: see text]-type -noise, including applications to animal models and possible impact on quantum foundations.

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54. Bias Reduction and Metric Learning for Nearest-Neighbor Estimation of Kullback-Leibler Divergence

Yung■Kyun Noh, Masashi Sugiyama, Song Liu, Marthinus Christoffel du Plessis, Frank C. Park, Daniel D. Lee
https://doi.org/10.1162/neco_a_01092 | pp. 1930-1960

Nearest-neighbor estimators for the Kullback-Leiber (KL) divergence that are asymptotically unbiased have recently been proposed and demonstrated in a number of applications. However, with a small number of samples, nonparametric methods typically suffer from large estimation bias due to the nonlocality of information derived from nearest-neighbor statistics. In this letter, we show that this estimation bias can be mitigated by modifying the metric function, and we propose a novel method for learning a locally optimal Mahalanobis distance function from parametric generative models of the underlying density distributions. Using both simulations and experiments on a variety of data sets, we demonstrate that this interplay between approximate generative models and nonparametric techniques can significantly improve the accuracy of nearest-neighbor-based estimation of the KL divergence.

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55. A Dynamic Neural Gradient Model of Two-Item and Intermediate Transposition

Gavin W. Jenkins, Paul Tupper
https://doi.org/10.1162/neco_a_01093 | pp. 1961-1982

Transposition is a tendency for organisms to generalize relationships between stimuli in situations where training does not objectively reward relationships over absolute, static associations. Transposition has most commonly been explained as either conceptual understanding of relationships (Köhler, 1938) as nonconceptual effects of neural memory gradients (as in Spence's stimulus discrimination theory, 1937). Most behavioral evidence can be explained by the gradient account, but a key finding unexplained by gradients is intermediate transposition, where a central (of three) stimulus, "relationally correct response," is generalized from training to test. Here, we introduce a dynamic neural field (DNF) model that captures intermediate transposition effects while using neural mechanisms closely resembling those of Spence's original proposal. The DNF model operates on dynamic rather than linear neural relationships, but it still functions by way of gradient interactions, and it does not invoke relational conceptual understanding in order to explain transposition behaviors. In addition to intermediate transposition, the DNF model also replicates the predictions of stimulus discrimination theory with respect to basic two-stimulus transposition. Effects of wider test item spacing were additionally captured. Overall, the DNF model captures a wider range of effects in transposition than stimulus discrimination theory, uses more fully specified neural mechanics, and integrates transposition into a wider modeling effort across cognitive tasks and phenomena. At the same time, the model features a similar low-level focus and emphasis on gradient interactions as Spence's, serving as a conceptual continuation and updating of Spence's work in the field of transposition.

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56. A Reinforcement Learning Neural Network for Robotic Manipulator Control

Yazhou Hu, Bailu Si
https://doi.org/10.1162/neco_a_01079 | pp. 1983-2004

We propose a neural network model for reinforcement learning to control a robotic manipulator with unknown parameters and dead zones. The model is composed of three networks. The state of the robotic manipulator is predicted by the state network of the model, the action policy is learned by the action network, and the performance index of the action policy is estimated by a critic network. The three networks work together to optimize the performance index based on the reinforcement learning control scheme. The convergence of the learning methods is analyzed. Application of the proposed model on a simulated two-link robotic manipulator demonstrates the effectiveness and the stability of the model.

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57. Why Does Large Batch Training Result in Poor Generalization? A Comprehensive Explanation and a Better Strategy from the Viewpoint of Stochastic Optimization

Tomoumi Takase, Satoshi Oyama, Masahito Kurihara
https://doi.org/10.1162/neco_a_01089 | pp. 2005-2023

We present a comprehensive framework of search methods, such as simulated annealing and batch training, for solving nonconvex optimization problems. These methods search a wider range by gradually decreasing the randomness added to the standard gradient descent method. The formulation that we define

on the basis of this framework can be directly applied to neural network training. This produces an effective approach that gradually increases batch size during training. We also explain why large batch training degrades generalization performance, which previous studies have not clarified.

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58. Machine Learning: Deepest Learning as Statistical Data Assimilation Problems

Henry D. I. Abarbanel, Paul J. Rozdeba, Sasha Shirman

https://doi.org/10.1162/neco_a_01094 | pp. 2025-2055

We formulate an equivalence between machine learning and the formulation of statistical data assimilation as used widely in physical and biological sciences. The correspondence is that layer number in a feedforward artificial network setting is the analog of time in the data assimilation setting. This connection has been noted in the machine learning literature. We add a perspective that expands on how methods from statistical physics and aspects of Lagrangian and Hamiltonian dynamics play a role in how networks can be trained and designed. Within the discussion of this equivalence, we show that adding more layers (making the network deeper) is analogous to adding temporal resolution in a data assimilation framework. Extending this equivalence to recurrent networks is also discussed. We explore how one can find a candidate for the global minimum of the cost functions in the machine learning context using a method from data assimilation. Calculations on simple models from both sides of the equivalence are reported. Also discussed is a framework in which the time or layer label is taken to be continuous, providing a differential equation, the Euler-Lagrange equation and its boundary conditions, as a necessary condition for a minimum of the cost function. This shows that the problem being solved is a two-point boundary value problem familiar in the discussion of variational methods. The use of continuous layers is denoted "deepest learning." These problems respect a symplectic symmetry in continuous layer phase space. Both Lagrangian versions and Hamiltonian versions of these problems are presented. Their well-studied implementation in a discrete time/layer, while respecting the symplectic structure, is addressed. The Hamiltonian version provides a direct rationale for backpropagation as a solution method for a certain two-point boundary value problem.

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59. Optimal Decoding of Dynamic Stimuli by Heterogeneous Populations of Spiking Neurons: A Closed-Form Approximation

Yuval Harel, Ron Meir, Manfred Opper

https://doi.org/10.1162/neco_a_01105 | pp. 2056-2112

Neural decoding may be formulated as dynamic state estimation (filtering) based on point-process observations, a generally intractable problem. Numerical sampling techniques are often practically useful for the decoding of real neural data. However, they are less useful as theoretical tools for modeling and understanding sensory neural systems, since they lead to limited conceptual insight into optimal encoding and decoding strategies. We consider sensory neural populations characterized by a distribution over neuron parameters. We develop an analytically tractable Bayesian approximation to optimal filtering based on the observation of spiking activity that greatly facilitates the analysis of optimal encoding in situations deviating from common assumptions of uniform coding. Continuous distributions are used to approximate large populations with few parameters, resulting in a filter whose complexity does not grow with population size and allowing optimization of population parameters rather than individual tuning functions. Numerical

comparison with particle filtering demonstrates the quality of the approximation. The analytic framework leads to insights that are difficult to obtain from numerical algorithms and is consistent with biological observations about the distribution of sensory cells' preferred stimuli.

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60. Neural Simpletrons: Learning in the Limit of Few Labels with Directed Generative Networks

Dennis Forster, Abdul-Saboor Sheikh, Jörg Lücke

https://doi.org/10.1162/neco_a_01100 | pp. 2113-2174

We explore classifier training for data sets with very few labels. We investigate this task using a neural network for nonnegative data. The network is derived from a hierarchical normalized Poisson mixture model with one observed and two hidden layers. With the single objective of likelihood optimization, both labeled and unlabeled data are naturally incorporated into learning. The neural activation and learning equations resulting from our derivation are concise and local. As a consequence, the network can be scaled using standard deep learning tools for parallelized GPU implementation. Using standard benchmarks for nonnegative data, such as text document representations, MNIST, and NIST SD19, we study the classification performance when very few labels are used for training. In different settings, the network's performance is compared to standard and recently suggested semisupervised classifiers. While other recent approaches are more competitive for many labels or fully labeled data sets, we find that the network studied here can be applied to numbers of few labels where no other system has been reported to operate so far.

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61. Methods for Assessment of Memory Reactivation

Shizhao Liu, Andres Grosmark, Zhe Chen

https://doi.org/10.1162/neco_a_01090 | pp. 2175-2209

It has been suggested that reactivation of previously acquired experiences or stored information in declarative memories in the hippocampus and neocortex contributes to memory consolidation and learning. Understanding memory consolidation depends crucially on the development of robust statistical methods for assessing memory reactivation. To date, several statistical methods have seen established for assessing memory reactivation based on bursts of ensemble neural spike activity during offline states. Using population-decoding methods, we propose a new statistical metric, the weighted distance correlation, to assess hippocampal memory reactivation (i.e., spatial memory replay) during quiet wakefulness and slow-wave sleep. The new metric can be combined with an unsupervised population decoding analysis, which is invariant to latent state labeling and allows us to detect statistical dependency beyond linearity in memory traces. We validate the new metric using two rat hippocampal recordings in spatial navigation tasks. Our proposed analysis framework may have a broader impact on assessing memory reactivations in other brain regions under different behavioral tasks.

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62. Evidence of Rentian Scaling of Functional Modules in Diverse Biological Networks

Javier J. How, Saket Navlakha

https://doi.org/10.1162/neco_a_01095 | pp. 2210-2244

Biological networks have long been known to be modular, containing sets of nodes that are highly connected internally. Less emphasis, however, has been placed on understanding how intermodule connections are distributed within a network. Here, we borrow ideas from engineered circuit design and study Rentian scaling, which states that the number of external connections between nodes in different modules is related to the number of nodes inside the modules by a power-law relationship. We tested this property in a broad class of molecular networks, including protein interaction networks for six species and gene regulatory networks for 41 human and 25 mouse cell types. Using evolutionarily defined modules corresponding to known biological processes in the cell, we found that all networks displayed Rentian scaling with a broad range of exponents. We also found evidence for Rentian scaling in functional modules in the Caenorhabditis elegans neural network, but, interestingly, not in three different social networks, suggesting that this property does not inevitably emerge. To understand how such scaling may have arisen evolutionarily, we derived a new graph model that can generate Rentian networks given a target Rent exponent and a module decomposition as inputs. Overall, our work uncovers a new principle shared by engineered circuits and biological networks.

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63. A Comparative Study of Pairwise Learning Methods Based on Kernel Ridge Regression

Michiel Stock, Tapio Pahikkala, Antti Airola, Bernard De Baets, Willem Waegeman

https://doi.org/10.1162/neco_a_01096 | pp. 2245-2283

Many machine learning problems can be formulated as predicting labels for a pair of objects. Problems of that kind are often referred to as pairwise learning, dyadic prediction, or network inference problems. During the past decade, kernel methods have played a dominant role in pairwise learning. They still obtain a state-of-the-art predictive performance, but a theoretical analysis of their behavior has been underexplored in the machine learning literature. In this work we review and unify kernel-based algorithms that are commonly used in different pairwise learning settings, ranging from matrix filtering to zero-shot learning. To this end, we focus on closed-form efficient instantiations of Kronecker kernel ridge regression. We show that independent task kernel ridge regression, two-step kernel ridge regression, and a linear matrix filter arise naturally as a special case of Kronecker kernel ridge regression, implying that all these methods implicitly minimize a squared loss. In addition, we analyze universality, consistency, and spectral filtering properties. Our theoretical results provide valuable insights into assessing the advantages and limitations of existing pairwise learning methods.

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64. Confounder Detection in High-Dimensional Linear Models Using First Moments of Spectral Measures

Furui Liu, Laiwan Chan

https://doi.org/10.1162/neco_a_01099 | pp. 2284-2318

In this letter, we study the confounder detection problem in the linear model, where the target variable [Formula: see text] is predicted using its [Formula: see text] potential causes [Formula: see text]. Based on an assumption of a rotation-invariant generating process of the model, recent study shows that the spectral

measure induced by the regression coefficient vector with respect to the covariance matrix of [Formula: see text] is close to a uniform measure in purely causal cases, but it differs from a uniform measure characteristically in the presence of a scalar confounder. Analyzing spectral measure patterns could help to detect confounding. In this letter, we propose to use the first moment of the spectral measure for confounder detection. We calculate the first moment of the regression vector-induced spectral measure and compare it with the first moment of a uniform spectral measure, both defined with respect to the covariance matrix of [Formula: see text]. The two moments coincide in nonconfounding cases and differ from each other in the presence of confounding. This statistical causal-confounding asymmetry can be used for confounder detection. Without the need to analyze the spectral measure pattern, our method avoids the difficulty of metric choice and multiple parameter optimization. Experiments on synthetic and real data show the performance of this method.

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65. The Discrete and Continuous Brain: From Decisions to Movement—And Back Again

Thomas Parr, Karl Friston

https://doi.org/10.1162/neco_a_01102 | pp. 2319-2347

To act upon the world, creatures must change continuous variables such as muscle length or chemical concentration. In contrast, decision making is an inherently discrete process, involving the selection among alternative courses of action. In this article, we consider the interface between the discrete and continuous processes that translate our decisions into movement in a Newtonian world-and how movement informs our decisions. We do so by appealing to active inference, with a special focus on the oculomotor system. Within this exemplar system, we argue that the superior colliculus is well placed to act as a discrete-continuous interface. Interestingly, when the neuronal computations within the superior colliculus are formulated in terms of active inference, we find that many aspects of its neuroanatomy emerge from the computations it must perform in this role.

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66. A New Foreperiod Effect on Intertrial Phase Coherence. Part I: Existence and Behavioral Relevance

Joaquín Rapela, Marissa Westerfield, Jeanne Townsend

https://doi.org/10.1162/neco_a_01109 | pp. 2348-2383

This letter makes scientific and methodological contributions. Scientifically, it demonstrates a new and behaviorally relevant effect of temporal expectation on the phase coherence of the electroencephalogram (EEG). Methodologically, it introduces novel methods to characterize EEG recordings at the single-trial level. Expecting events in time can lead to more efficient behavior. A remarkable finding in the study of temporal expectation is the foreperiod effect on reaction time, that is, the influence on reaction time of the delay between a warning signal and a succeeding imperative stimulus to which subjects are instructed to respond as quickly as possible. Here we study a new foreperiod effect in an audiovisual attention-shifting oddball task in which attention-shift cues directed the attention of subjects to impendent deviant stimuli of a given modality and therefore acted as warning signals for these deviants. Standard stimuli, to which subjects did not respond, were interspersed between warning signals and deviants. We hypothesized that foreperiod durations modulated intertrial phase coherence (ITPC, the degree of phase alignment across multiple trials)

evoked by behaviorally irrelevant standards and that these modulations are behaviorally meaningful. Using averaged data, we first observed that ITPC evoked by standards closer to the warning signal was significantly different from that evoked by standards further away from it, establishing a new foreperiod effect on ITPC evoked by standards. We call this effect the standard foreperiod (SFP) effect on ITPC. We reasoned that if the SFP influences ITPC evoked by standards, it should be possible to decode the former from the latter on a trial-by-trial basis. We were able to do so showing that this effect can be observed in single trials. We demonstrated the behavioral relevance of the SFP effect on ITPC by showing significant correlations between its strength and subjects' behavioral performance.

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67. Spiking Elementary Motion Detector in Neuromorphic Systems

Moritz B. Milde, Olivier J. N. Bertrand, Hemalatha Ramachandran, Martin Egelhaaf, Elisabetta Chicca
https://doi.org/10.1162/neco_a_01112 | pp. 2384-2417

Apparent motion of the surroundings on an agent's retina can be used to navigate through cluttered environments, avoid collisions with obstacles, or track targets of interest. The pattern of apparent motion of objects, (i.e., the optic flow), contains spatial information about the surrounding environment. For a small, fast-moving agent, as used in search and rescue missions, it is crucial to estimate the distance to close-by objects to avoid collisions quickly. This estimation cannot be done by conventional methods, such as frame-based optic flow estimation, given the size, power, and latency constraints of the necessary hardware. A practical alternative makes use of event-based vision sensors. Contrary to the frame-based approach, they produce so-called events only when there are changes in the visual scene. We propose a novel asynchronous circuit, the spiking elementary motion detector (sEMD), composed of a single silicon neuron and synapse, to detect elementary motion from an event-based vision sensor. The sEMD encodes the time an object's image needs to travel across the retina into a burst of spikes. The number of spikes within the burst is proportional to the speed of events across the retina. A fast but imprecise estimate of the time-to-travel can already be obtained from the first two spikes of a burst and refined by subsequent interspike intervals. The latter encoding scheme is possible due to an adaptive nonlinear synaptic efficacy scaling. We show that the sEMD can be used to compute a collision avoidance direction in the context of robotic navigation in a cluttered outdoor environment and compared the collision avoidance direction to a frame-based algorithm. The proposed computational principle constitutes a generic spiking temporal correlation detector that can be applied to other sensory modalities (e.g., sound localization), and it provides a novel perspective to gating information in spiking neural networks.

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68. A Basic Phase Diagram of Neuronal Dynamics

Wenyuan Li, I. V. Ovchinnikov, Honglin Chen, Zhe Wang, Albert Lee, Houchul Lee, Carlos Cepeda, Robert N. Schwartz, Karlheinz Meier, Kang L. Wang
https://doi.org/10.1162/neco_a_01103 | pp. 2418-2438

The extreme complexity of the brain has attracted the attention of neuroscientists and other researchers for a long time. More recently, the neuromorphic hardware has matured to provide a new powerful tool to study neuronal dynamics. Here, we study neuronal dynamics using different settings on a neuromorphic chip built with flexible parameters of neuron models. Our unique setting in the network of leaky integrate-and-fire (LIF) neurons is to introduce a weak noise environment. We observed three different types of collective neuronal

activities, or phases, separated by sharp boundaries, or phase transitions. From this, we construct a rudimentary phase diagram of neuronal dynamics and demonstrate that a noise-induced chaotic phase (N-phase), which is dominated by neuronal avalanche activity (intermittent aperiodic neuron firing), emerges in the presence of noise and its width grows with the noise intensity. The dynamics can be manipulated in this N-phase. Our results and comparison with clinical data is consistent with the literature and our previous work showing that healthy brain must reside in the N-phase. We argue that the brain phase diagram with further refinement may be used for the diagnosis and treatment of mental disease and also suggest that the dynamics may be manipulated to serve as a means of new information processing (e.g., for optimization). Neuromorphic chips, similar to the one we used but with a variety of neuron models, may be used to further enhance the understanding of human brain function and accelerate the development of neuroscience research.

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69. A Computationally Efficient Visual Saliency Algorithm Suitable for an Analog CMOS Implementation

Robert D'Angelo, Richard J. Wood, Nathan Lowry, Geremy Freifeld, Haiyao Huang, Christopher Salthouse, Brent Hollosi, Matthew Muresan, Wes Uy, Nhut Tran, Armand Chery, Dorothy C. Poppe, Sameer Sonkusale

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Computer vision algorithms are often limited in their application by the large amount of data that must be processed. Mammalian vision systems mitigate this high bandwidth requirement by prioritizing certain regions of the visual field with neural circuits that select the most salient regions. This work introduces a novel and computationally efficient visual saliency algorithm for performing this neuromorphic attention-based data reduction. The proposed algorithm has the added advantage that it is compatible with an analog CMOS design while still achieving comparable performance to existing state-of-the-art saliency algorithms. This compatibility allows for direct integration with the analog-to-digital conversion circuitry present in CMOS image sensors. This integration leads to power savings in the converter by quantizing only the salient pixels. Further system-level power savings are gained by reducing the amount of data that must be transmitted and processed in the digital domain. The analog CMOS compatible formulation relies on a pulse width (i.e., time mode) encoding of the pixel data that is compatible with pulse-mode imagers and slope based converters often used in imager designs. This letter begins by discussing this time-mode encoding for implementing neuromorphic architectures. Next, the proposed algorithm is derived. Hardware-oriented optimizations and modifications to this algorithm are proposed and discussed. Next, a metric for quantifying saliency accuracy is proposed, and simulation results of this metric are presented. Finally, an analog synthesis approach for a time-mode architecture is outlined, and postsynthesis transistor-level simulations that demonstrate functionality of an implementation in a modern CMOS process are discussed.

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70. ASIC Implementation of a Nonlinear Dynamical Model for Hippocampal Prosthesis

Zhi-Tong Qiao, Han Yan, Xiaoxia Han, Xu Han, Xiangyu Li, Dong Song, Theodore W. Berger, Ray C. C. Cheung

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A hippocampal prosthesis is a very large scale integration (VLSI) biochip that needs to be implanted in the biological brain to solve a cognitive dysfunction. In this letter, we propose a novel low-complexity, small-area, and low-power programmable hippocampal neural network application-specific integrated circuit (ASIC) for a hippocampal prosthesis. It is based on the nonlinear dynamical model of the hippocampus: namely multi-input, multi-output (MIMO)-generalized Laguerre-Volterra model (GLVM). It can realize the real-time prediction of hippocampal neural activity. New hardware architecture, a storage space configuration scheme, low-power convolution, and gaussian random number generator modules are proposed. The ASIC is fabricated in 40 nm technology with a core area of 0.122 mm[Formula: see text] and test power of 84.4 [Formula: see text]W. Compared with the design based on the traditional architecture, experimental results show that the core area of the chip is reduced by 84.94% and the core power is reduced by 24.30%.

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71. A Locally Optimal Algorithm for Estimating a Generating Partition from an Observed Time Series and Its Application to Anomaly Detection

Najah F. Ghalyan, David J. Miller, Asok Ray

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Estimation of a generating partition is critical for symbolization of measurements from discrete-time dynamical systems, where a sequence of symbols from a (finite-cardinality) alphabet may uniquely specify the underlying time series. Such symbolization is useful for computing measures (e.g., Kolmogorov-Sinai entropy) to identify or characterize the (possibly unknown) dynamical system. It is also useful for time series classification and anomaly detection. The seminal work of Hirata, Judd, and Kilminster (2004) derives a novel objective function, akin to a clustering objective, that measures the discrepancy between a set of reconstruction values and the points from the time series. They cast estimation of a generating partition via the minimization of their objective function. Unfortunately, their proposed algorithm is nonconvergent, with no guarantee of finding even locally optimal solutions with respect to their objective. The difficulty is a heuristic nearest neighbor symbol assignment step. Alternatively, we develop a novel, locally optimal algorithm for their objective. We apply iterative nearest-neighbor symbol assignments with guaranteed discrepancy descent, by which joint, locally optimal symbolization of the entire time series is achieved. While most previous approaches frame generating partition estimation as a state-space partitioning problem, we recognize that minimizing the Hirata et al. (2004) objective function does not induce an explicit partitioning of the state space, but rather the space consisting of the entire time series (effectively, clustering in a (countably) infinite-dimensional space). Our approach also amounts to a novel type of sliding block lossy source coding. Improvement, with respect to several measures, is demonstrated over popular methods for symbolizing chaotic maps. We also apply our approach to time-series anomaly detection, considering both chaotic maps and failure application in a polycrystalline alloy material.

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72. Active Inference, Belief Propagation, and the Bethe Approximation

Sarah Schwöbel, Stefan J. Kiebel, Dimitrije Markovi■

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When modeling goal-directed behavior in the presence of various sources of uncertainty, planning can be described as an inference process. A solution to the problem of planning as inference was previously

proposed in the active inference framework in the form of an approximate inference scheme based on variational free energy. However, this approximate scheme was based on the mean-field approximation, which assumes statistical independence of hidden variables and is known to show overconfidence and may converge to local minima of the free energy. To better capture the spatiotemporal properties of an environment, we reformulated the approximate inference process using the so-called Bethe approximation. Importantly, the Bethe approximation allows for representation of pairwise statistical dependencies. Under these assumptions, the minimizer of the variational free energy corresponds to the belief propagation algorithm, commonly used in machine learning. To illustrate the differences between the mean-field approximation and the Bethe approximation, we have simulated agent behavior in a simple goal-reaching task with different types of uncertainties. Overall, the Bethe agent achieves higher success rates in reaching goal states. We relate the better performance of the Bethe agent to more accurate predictions about the consequences of its own actions. Consequently, active inference based on the Bethe approximation extends the application range of active inference to more complex behavioral tasks.

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73. An Empirical Evaluation of Rule Extraction from Recurrent Neural Networks

Qinglong Wang, Kaixuan Zhang, Alexander G. Ororbia, Xinyu Xing, Xue Liu, C. Lee Giles
https://doi.org/10.1162/neco_a_01111 | pp. 2568-2591

Rule extraction from black box models is critical in domains that require model validation before implementation, as can be the case in credit scoring and medical diagnosis. Though already a challenging problem in statistical learning in general, the difficulty is even greater when highly nonlinear, recursive models, such as recurrent neural networks (RNNs), are fit to data. Here, we study the extraction of rules from second-order RNNs trained to recognize the Tomita grammars. We show that production rules can be stably extracted from trained RNNs and that in certain cases, the rules outperform the trained RNNs.

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74. Learning Data Manifolds with a Cutting Plane Method

SueYeon Chung, Uri Cohen, Haim Sompolinsky, Daniel D. Lee
https://doi.org/10.1162/neco_a_01119 | pp. 2593-2615

We consider the problem of classifying data manifolds where each manifold represents invariances that are parameterized by continuous degrees of freedom. Conventional data augmentation methods rely on sampling large numbers of training examples from these manifolds. Instead, we propose an iterative algorithm, [Formula: see text], based on a cutting plane approach that efficiently solves a quadratic semi-infinite programming problem to find the maximum margin solution. We provide a proof of convergence as well as a polynomial bound on the number of iterations required for a desired tolerance in the objective function. The efficiency and performance of [Formula: see text] are demonstrated in high-dimensional simulations and on image manifolds generated from the ImageNet data set. Our results indicate that [Formula: see text] is able to rapidly learn good classifiers and shows superior generalization performance compared with conventional maximum margin methods using data augmentation methods.

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75. Recognition Dynamics in the Brain under the Free Energy Principle

Chang Sub Kim

https://doi.org/10.1162/neco_a_01115 | pp. 2616-2659

We formulate the computational processes of perception in the framework of the principle of least action by postulating the theoretical action as a time integral of the variational free energy in the neurosciences. The free energy principle is accordingly rephrased, on autopoetic grounds, as follows: all viable organisms attempt to minimize their sensory uncertainty about an unpredictable environment over a temporal horizon. By taking the variation of informational action, we derive neural recognition dynamics (RD), which by construction reduces to the Bayesian filtering of external states from noisy sensory inputs. Consequently, we effectively cast the gradient-descent scheme of minimizing the free energy into Hamiltonian mechanics by addressing only the positions and momenta of the organisms' representations of the causal environment. To demonstrate the utility of our theory, we show how the RD may be implemented in a neuronally based biophysical model at a single-cell level and subsequently in a coarse-grained, hierarchical architecture of the brain. We also present numerical solutions to the RD for a model brain and analyze the perceptual trajectories around attractors in neural state space.

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76. Computing with Spikes: The Advantage of Fine-Grained Timing

Stephen Verzi, Fredrick Rothganger, Ojas Parekh, Tu-Thach Quach, Nadine E. Miner, Craig M. Vineyard, Conrad D. James, James B. Aimone

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Neural-inspired spike-based computing machines often claim to achieve considerable advantages in terms of energy and time efficiency by using spikes for computation and communication. However, fundamental questions about spike-based computation remain unanswered. For instance, how much advantage do spike-based approaches have over conventional methods, and under what circumstances does spike-based computing provide a comparative advantage? Simply implementing existing algorithms using spikes as the medium of computation and communication is not guaranteed to yield an advantage. Here, we demonstrate that spike-based communication and computation within algorithms can increase throughput, and they can decrease energy cost in some cases. We present several spiking algorithms, including sorting a set of numbers in ascending/descending order, as well as finding the maximum or minimum or median of a set of numbers. We also provide an example application: a spiking median-filtering approach for image processing providing a low-energy, parallel implementation. The algorithms and analyses presented here demonstrate that spiking algorithms can provide performance advantages and offer efficient computation of fundamental operations useful in more complex algorithms.

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77. Hexagonal Grid Fields Optimally Encode Transitions in Spatiotemporal Sequences

Nicolai Waniek

https://doi.org/10.1162/neco_a_01122 | pp. 2691-2725

Grid cells of the rodent entorhinal cortex are essential for spatial navigation. Although their function is commonly believed to be either path integration or localization, the origin or purpose of their hexagonal firing

fields remains disputed. Here they are proposed to arise as an optimal encoding of transitions in sequences. First, storage requirements for transitions in general episodic sequences are examined using propositional logic and graph theory. Subsequently, transitions in complete metric spaces are considered under the assumption of an ideal sampling of an input space. It is shown that memory capacity of neurons that have to encode multiple feasible spatial transitions is maximized by a hexagonal pattern. Grid cells are proposed to encode spatial transitions in spatiotemporal sequences, with the entorhinal-hippocampal loop forming a multitransition system.

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78. CosMIC: A Consistent Metric for Spike Inference from Calcium Imaging

Stephanie Reynolds, Therése Abrahamsson, P. Jesper Sjöström, Simon R. Schultz, Pier Luigi Dragotti
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In recent years, the development of algorithms to detect neuronal spiking activity from two-photon calcium imaging data has received much attention, yet few researchers have examined the metrics used to assess the similarity of detected spike trains with the ground truth. We highlight the limitations of the two most commonly used metrics, the spike train correlation and success rate, and propose an alternative, which we refer to as CosMIC. Rather than operating on the true and estimated spike trains directly, the proposed metric assesses the similarity of the pulse trains obtained from convolution of the spike trains with a smoothing pulse. The pulse width, which is derived from the statistics of the imaging data, reflects the temporal tolerance of the metric. The final metric score is the size of the commonalities of the pulse trains as a fraction of their average size. Viewed through the lens of set theory, CosMIC resembles a continuous Sørensen-Dice coefficient-an index commonly used to assess the similarity of discrete, presence/absence data. We demonstrate the ability of the proposed metric to discriminate the precision and recall of spike train estimates. Unlike the spike train correlation, which appears to reward overestimation, the proposed metric score is maximized when the correct number of spikes have been detected. Furthermore, we show that CosMIC is more sensitive to the temporal precision of estimates than the success rate.

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79. Autoregressive Point Processes as Latent State-Space Models: A Moment-Closure Approach to Fluctuations and Autocorrelations

Michael Rule, Guido Sanguinetti
https://doi.org/10.1162/neco_a_01121 | pp. 2757-2780

Modeling and interpreting spike train data is a task of central importance in computational neuroscience, with significant translational implications. Two popular classes of data-driven models for this task are autoregressive point-process generalized linear models (PPGLM) and latent state-space models (SSM) with point-process observations. In this letter, we derive a mathematical connection between these two classes of models. By introducing an auxiliary history process, we represent exactly a PPGLM in terms of a latent, infinite-dimensional dynamical system, which can then be mapped onto an SSM by basis function projections and moment closure. This representation provides a new perspective on widely used methods for modeling spike data and also suggests novel algorithmic approaches to fitting such models. We illustrate our results on a phasic bursting neuron model, showing that our proposed approach provides an accurate and efficient way to capture neural dynamics.

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80. Multiclass Classification and Feature Selection Based on Least Squares Regression with Large Margin

Haifeng Zhao, Siqi Wang, Zheng Wang

https://doi.org/10.1162/neco_a_01116 | pp. 2781-2804

Least squares regression (LSR) is a fundamental statistical analysis technique that has been widely applied to feature learning. However, limited by its simplicity, the local structure of data is easy to neglect, and many methods have considered using orthogonal constraint for preserving more local information. Another major drawback of LSR is that the loss function between soft regression results and hard target values cannot precisely reflect the classification ability; thus, the idea of the large margin constraint is put forward. As a consequence, we pay attention to the concepts of large margin and orthogonal constraint to propose a novel algorithm, orthogonal least squares regression with large margin (OLSLM), for multiclass classification in this letter. The core task of this algorithm is to learn regression targets from data and an orthogonal transformation matrix simultaneously such that the proposed model not only ensures every data point can be correctly classified with a large margin than conventional least squares regression, but also can preserve more local data structure information in the subspace. Our efficient optimization method for solving the large margin constraint and orthogonal constraint iteratively proved to be convergent in both theory and practice. We also apply the large margin constraint in the process of generating a sparse learning model for feature selection via joint [Formula: see text]-norm minimization on both loss function and regularization terms. Experimental results validate that our method performs better than state-of-the-art methods on various real-world data sets.

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81. Adaptive Learning Algorithm Convergence in Passive and Reactive Environments

Richard M. Golden

https://doi.org/10.1162/neco_a_01117 | pp. 2805-2832

Although the number of artificial neural network and machine learning architectures is growing at an exponential pace, more attention needs to be paid to theoretical guarantees of asymptotic convergence for novel, nonlinear, high-dimensional adaptive learning algorithms. When properly understood, such guarantees can guide the algorithm development and evaluation process and provide theoretical validation for a particular algorithm design. For many decades, the machine learning community has widely recognized the importance of stochastic approximation theory as a powerful tool for identifying explicit convergence conditions for adaptive learning machines. However, the verification of such conditions is challenging for multidisciplinary researchers not working in the area of stochastic approximation theory. For this reason, this letter presents a new stochastic approximation theorem for both passive and reactive learning environments with assumptions that are easily verifiable. The theorem is widely applicable to the analysis and design of important machine learning algorithms including deep learning algorithms with multiple strict local minimizers, Monte Carlo expectation-maximization algorithms, contrastive divergence learning in Markov fields, and policy gradient reinforcement learning.

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82. Improving Stock Closing Price Prediction Using Recurrent Neural Network and Technical Indicators

Tingwei Gao, Yueting Chai

https://doi.org/10.1162/neco_a_01124 | pp. 2833-2854

This study focuses on predicting stock closing prices by using recurrent neural networks (RNNs). A long short-term memory (LSTM) model, a type of RNN coupled with stock basic trading data and technical indicators, is introduced as a novel method to predict the closing price of the stock market. We realize dimension reduction for the technical indicators by conducting principal component analysis (PCA). To train the model, some optimization strategies are followed, including adaptive moment estimation (Adam) and Glorot uniform initialization. Case studies are conducted on Standard & Poor's 500, NASDAQ, and Apple (AAPL). Plenty of comparison experiments are performed using a series of evaluation criteria to evaluate this model. Accurate prediction of stock market is considered an extremely challenging task because of the noisy environment and high volatility associated with the external factors. We hope the methodology we propose advances the research for analyzing and predicting stock time series. As the results of experiments suggest, the proposed model achieves a good level of fitness.

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83. Applications of Recurrent Neural Networks in Environmental Factor Forecasting: A Review

Yingyi Chen, Qianqian Cheng, Yanjun Cheng, Hao Yang, Huihui Yu

https://doi.org/10.1162/neco_a_01134 | pp. 2855-2881

Analysis and forecasting of sequential data, key problems in various domains of engineering and science, have attracted the attention of many researchers from different communities. When predicting the future probability of events using time series, recurrent neural networks (RNNs) are an effective tool that have the learning ability of feedforward neural networks and expand their expression ability using dynamic equations. Moreover, RNNs are able to model several computational structures. Researchers have developed various RNNs with different architectures and topologies. To summarize the work of RNNs in forecasting and provide guidelines for modeling and novel applications in future studies, this review focuses on applications of RNNs for time series forecasting in environmental factor forecasting. We present the structure, processing flow, and advantages of RNNs and analyze the applications of various RNNs in time series forecasting. In addition, we discuss limitations and challenges of applications based on RNNs and future research directions. Finally, we summarize applications of RNNs in forecasting.

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84. Diplomats' Mystery Illness and Pulsed Radiofrequency/Microwave Radiation

Beatrice A. Golomb

https://doi.org/10.1162/neco_a_01133 | pp. 2882-2985

Abstract Importance: A mystery illness striking U.S. and Canadian diplomats to Cuba (and now China) “has confounded the FBI, the State Department and US intelligence agencies” (Lederman, Weissenstein, & Lee, 2017). Sonic explanations for the so-called health attacks have long dominated media reports, propelled by peculiar sounds heard and auditory symptoms experienced. Sonic mediation was justly

rejected by experts. We assessed whether pulsed radiofrequency/microwave radiation (RF/MW) exposure can accommodate reported facts in diplomats, including unusual ones. Observations: (1) Noises: Many diplomats heard chirping, ringing or grinding noises at night during episodes reportedly triggering health problems. Some reported that noises were localized with laser-like precision or said the sounds seemed to follow them (within the territory in which they were perceived). Pulsed RF/MW engenders just these apparent “sounds” via the Frey effect. Perceived “sounds” differ by head dimensions and pulse characteristics and can be perceived as located behind in or above the head. Ability to hear the “sounds” depends on high-frequency hearing and low ambient noise. (2) Signs/symptoms: Hearing loss and tinnitus are prominent in affected diplomats and in RF/MW-affected individuals. Each of the protean symptoms that diplomats report also affect persons reporting symptoms from RF/MW: sleep problems, headaches, and cognitive problems dominate in both groups. Sensations of pressure or vibration figure in each. Both encompass vision, balance, and speech problems and nosebleeds. Brain injury and brain swelling are reported in both. (3) Mechanisms: Oxidative stress provides a documented mechanism of RF/MW injury compatible with reported signs and symptoms; sequelae of endothelial dysfunction (yielding blood flow compromise), membrane damage, blood-brain barrier disruption, mitochondrial injury, apoptosis, and autoimmune triggering afford downstream mechanisms, of varying persistence, that merit investigation. (4) Of note, microwaving of the U.S. embassy in Moscow is historically documented. Conclusions and relevance: Reported facts appear consistent with pulsed RF/MW as the source of injury in affected diplomats. Nondiplomats citing symptoms from RF/MW, often with an inciting pulsed-RF/MW exposure, report compatible health conditions. Under the RF/MW hypothesis, lessons learned for diplomats and for RF/MW-affected civilians may each aid the other.

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85. Robust Closed-Loop Control of a Cursor in a Person with Tetraplegia using Gaussian Process Regression

David M. Brandman, Michael C. Burkhart, Jessica N. Kelemen, Brian Franco, Matthew Tom Harrison, Leigh R. Hochberg

https://doi.org/10.1162/neco_a_01129 | pp. 2986-3008

Intracortical brain computer interfaces can enable individuals with paralysis to control external devices through voluntarily modulated brain activity. Decoding quality has been previously shown to degrade with signal nonstationarities-specifically, the changes in the statistics of the data between training and testing data sets. This includes changes to the neural tuning profiles and baseline shifts in firing rates of recorded neurons, as well as nonphysiological noise. While progress has been made toward providing long-term user control via decoder recalibration, relatively little work has been dedicated to making the decoding algorithm more resilient to signal nonstationarities. Here, we describe how principled kernel selection with gaussian process regression can be used within a Bayesian filtering framework to mitigate the effects of commonly encountered nonstationarities. Given a supervised training set of (neural features, intention to move in a direction)-pairs, we use gaussian process regression to predict the intention given the neural data. We apply kernel embedding for each neural feature with the standard radial basis function. The multiple kernels are then summed together across each neural dimension, which allows the kernel to effectively ignore large differences that occur only in a single feature. The summed kernel is used for real-time predictions of the posterior mean and variance under a gaussian process framework. The predictions are then filtered using the discriminative Kalman filter to produce an estimate of the neural intention given the history of neural data. We refer to the multiple kernel approach combined with the discriminative Kalman filter as the

MK-DKF. We found that the MK-DKF decoder was more resilient to nonstationarities frequently encountered in-real world settings yet provided similar performance to the currently used Kalman decoder. These results demonstrate a method by which neural decoding can be made more resistant to nonstationarities.

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86. A Simple Model for Low Variability in Neural Spike Trains

Ulisse Ferrari, Stéphane Deny, Olivier Marre, Thierry Mora

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Neural noise sets a limit to information transmission in sensory systems. In several areas, the spiking response (to a repeated stimulus) has shown a higher degree of regularity than predicted by a Poisson process. However, a simple model to explain this low variability is still lacking. Here we introduce a new model, with a correction to Poisson statistics, that can accurately predict the regularity of neural spike trains in response to a repeated stimulus. The model has only two parameters but can reproduce the observed variability in retinal recordings in various conditions. We show analytically why this approximation can work. In a model of the spike-emitting process where a refractory period is assumed, we derive that our simple correction can well approximate the spike train statistics over a broad range of firing rates. Our model can be easily plugged to stimulus processing models, like a linear-nonlinear model or its generalizations, to replace the Poisson spike train hypothesis that is commonly assumed. It estimates the amount of information transmitted much more accurately than Poisson models in retinal recordings. Thanks to its simplicity, this model has the potential to explain low variability in other areas.

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87. Circuit Polarity Effect of Cortical Connectivity, Activity, and Memory

Yoram Baram

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Experimental constraints have traditionally implied separate studies of different cortical functions, such as memory and sensory-motor control. Yet certain cortical modalities, while repeatedly observed and reported, have not been clearly identified with one cortical function or another. Specifically, while neuronal membrane and synapse polarities with respect to a certain potential value have been attracting considerable interest in recent years, the purposes of such polarities have largely remained a subject for speculation and debate. Formally identifying these polarities as on-off neuronal polarity gates, we analytically show that cortical circuit structure, behavior, and memory are all governed by the combined potent effect of these gates, which we collectively term circuit polarity. Employing widely accepted and biologically validated firing rate and plasticity paradigms, we show that circuit polarity is mathematically embedded in the corresponding models. Moreover, we show that the firing rate dynamics implied by these models are driven by ongoing circuit polarity gating dynamics. Furthermore, circuit polarity is shown to segregate cortical circuits into internally synchronous, externally asynchronous subcircuits, defining their firing rate modes in accordance with different cortical tasks. In contrast to the Hebbian paradigm, which is shown to be susceptible to mutual neuronal interference in the face of asynchrony, circuit polarity is shown to block such interference. Noting convergence of synaptic weights, we show that circuit polarity holds the key to cortical memory, having a segregated capacity linear in the number of neurons. While memory concealment is implied by complete neuronal silencing, memory is restored by reactivating the original circuit polarity. Finally, we show that incomplete deterioration or restoration of circuit polarity results in memory modification, which may be

associated with partial or false recall, or novel innovation.

88. Adaptive Gaussian Process Approximation for Bayesian Inference with Expensive Likelihood Functions

Hongqiao Wang, Jinglai Li

https://doi.org/10.1162/neco_a_01127 | pp. 3072-3094

We consider Bayesian inference problems with computationally intensive likelihood functions. We propose a Gaussian process (GP)-based method to approximate the joint distribution of the unknown parameters and the data, built on recent work (Kandasamy, Schneider, & Póczos, 2015). In particular, we write the joint density approximately as a product of an approximate posterior density and an exponentiated GP surrogate. We then provide an adaptive algorithm to construct such an approximation, where an active learning method is used to choose the design points. With numerical examples, we illustrate that the proposed method has competitive performance against existing approaches for Bayesian computation.

89. Convex Coupled Matrix and Tensor Completion

Kishan Wimalawarne, Makoto Yamada, Hiroshi Mamitsuka

https://doi.org/10.1162/neco_a_01123 | pp. 3095-3127

We propose a set of convex low-rank inducing norms for coupled matrices and tensors (hereafter referred to as coupled tensors), in which information is shared between the matrices and tensors through common modes. More specifically, we first propose a mixture of the overlapped trace norm and the latent norms with the matrix trace norm, and then, propose a completion model regularized using these norms to impute coupled tensors. A key advantage of the proposed norms is that they are convex and can be used to find a globally optimal solution, whereas existing methods for coupled learning are nonconvex. We also analyze the excess risk bounds of the completion model regularized using our proposed norms and show that they can exploit the low-rankness of coupled tensors, leading to better bounds compared to those obtained using uncoupled norms. Through synthetic and real-data experiments, we show that the proposed completion model compares favorably with existing ones.

90. Cross-Entropy Pruning for Compressing Convolutional Neural Networks

Rongxin Bao, Xu Yuan, Zhikui Chen, Ruixin Ma

https://doi.org/10.1162/neco_a_01131 | pp. 3128-3149

The success of CNNs is accompanied by deep models and heavy storage costs. For compressing CNNs, we propose an efficient and robust pruning approach, cross-entropy pruning (CEP). Given a trained CNN model, connections were divided into groups in a group-wise way according to their corresponding output neurons. All connections with their cross-entropy errors below a grouping threshold were then removed. A sparse model was obtained and the number of parameters in the baseline model significantly reduced. This letter also presents a highest cross-entropy pruning (HCEP) method that keeps a small portion of weights with the highest CEP. This method further improves the accuracy of CEP. To validate CEP, we conducted the experiments on low redundant networks that are hard to compress. For the MNIST data set, CEP

achieves an 0.08% accuracy drop required by LeNet-5 benchmark with only 16% of original parameters. Our proposed CEP also reduces approximately 75% of the storage cost of AlexNet on the ILSVRC 2012 data set, increasing the top-1 errorby only 0.4% and top-5 error by only 0.2%. Compared with three existing methods on LeNet-5, our proposed CEP and HCEP perform significantly better than the existing methods in terms of the accuracy and stability. Some computer vision tasks on CNNs such as object detection and style transfer can be computed in a high-performance way using our CEP and HCEP strategies.

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91. Dense Associative Memory Is Robust to Adversarial Inputs

Dmitry Krotov, John Hopfield

https://doi.org/10.1162/neco_a_01143 | pp. 3151-3167

Deep neural networks (DNNs) trained in a supervised way suffer from two known problems. First, the minima of the objective function used in learning correspond to data points (also known as rubbish examples or fooling images) that lack semantic similarity with the training data. Second, a clean input can be changed by a small, and often imperceptible for human vision, perturbation so that the resulting deformed input is misclassified by the network. These findings emphasize the differences between the ways DNNs and humans classify patterns and raise a question of designing learning algorithms that more accurately mimic human perception compared to the existing methods. Our article examines these questions within the framework of dense associative memory (DAM) models. These models are defined by the energy function, with higher-order (higher than quadratic) interactions between the neurons. We show that in the limit when the power of the interaction vertex in the energy function is sufficiently large, these models have the following three properties. First, the minima of the objective function are free from rubbish images, so that each minimum is a semantically meaningful pattern. Second, artificial patterns poised precisely at the decision boundary look ambiguous to human subjects and share aspects of both classes that are separated by that decision boundary. Third, adversarial images constructed by models with small power of the interaction vertex, which are equivalent to DNN with rectified linear units, fail to transfer to and fool the models with higher-order interactions. This opens up the possibility of using higher-order models for detecting and stopping malicious adversarial attacks. The results we present suggest that DAMs with higher-order energy functions are more robust to adversarial and rubbish inputs than DNNs with rectified linear units.

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92. Unconscious Biases in Neural Populations Coding Multiple Stimuli

Sander Keemink, Dharmesh Tailor, Mark C. W. van Rossum

https://doi.org/10.1162/neco_a_01130 | pp. 3168-3188

Throughout the nervous system, information is commonly coded in activity distributed over populations of neurons. In idealized situations where a single, continuous stimulus is encoded in a homogeneous population code, the value of the encoded stimulus can be read out without bias. However, in many situations, multiple stimuli are simultaneously present; for example, multiple motion patterns might overlap. Here we find that when multiple stimuli that overlap in their neural representation are simultaneously encoded in the population, biases in the read-out emerge. Although the bias disappears in the absence of noise, the bias is remarkably persistent at low noise levels. The bias can be reduced by competitive encoding schemes or by employing complex decoders. To study the origin of the bias, we develop a novel

general framework based on gaussian processes that allows an accurate calculation of the estimate distributions of maximum likelihood decoders, and reveals that the distribution of estimates is bimodal for overlapping stimuli. The results have implications for neural coding and behavioral experiments on, for instance, overlapping motion patterns.

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93. Nonlinear Modeling of Neural Interaction for Spike Prediction Using the Staged Point-Process Model

Cunle Qian, Xuyun Sun, Shaomin Zhang, Dong Xing, Hongbao Li, Xiaoxiang Zheng, Gang Pan, Yiwen Wang

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Neurons communicate nonlinearly through spike activities. Generalized linear models (GLMs) describe spike activities with a cascade of a linear combination across inputs, a static nonlinear function, and an inhomogeneous Bernoulli or Poisson process, or Cox process if a self-history term is considered. This structure considers the output nonlinearity in spike generation but excludes the nonlinear interaction among input neurons. Recent studies extend GLMs by modeling the interaction among input neurons with a quadratic function, which considers the interaction between every pair of input spikes. However, quadratic effects may not fully capture the nonlinear nature of input interaction. We therefore propose a staged point-process model to describe the nonlinear interaction among inputs using a few hidden units, which follows the idea of artificial neural networks. The output firing probability conditioned on inputs is formed as a cascade of two linear-nonlinear (a linear combination plus a static nonlinear function) stages and an inhomogeneous Bernoulli process. Parameters of this model are estimated by maximizing the log likelihood on output spike trains. Unlike the iterative reweighted least squares algorithm used in GLMs, where the performance is guaranteed by the concave condition, we propose a modified Levenberg-Marquardt (L-M) algorithm, which directly calculates the Hessian matrix of the log likelihood, for the nonlinear optimization in our model. The proposed model is tested on both synthetic data and real spike train data recorded from the dorsal premotor cortex and primary motor cortex of a monkey performing a center-out task. Performances are evaluated by discrete-time rescaled Kolmogorov-Smirnov tests, where our model statistically outperforms a GLM and its quadratic extension, with a higher goodness-of-fit in the prediction results. In addition, the staged point-process model describes nonlinear interaction among input neurons with fewer parameters than quadratic models, and the modified L-M algorithm also demonstrates fast convergence.

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94. Omitted Variable Bias in GLMs of Neural Spiking Activity

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Generalized linear models (GLMs) have a wide range of applications in systems neuroscience describing the encoding of stimulus and behavioral variables, as well as the dynamics of single neurons. However, in any given experiment, many variables that have an impact on neural activity are not observed or not modeled. Here we demonstrate, in both theory and practice, how these omitted variables can result in biased parameter estimates for the effects that are included. In three case studies, we estimate tuning functions for common experiments in motor cortex, hippocampus, and visual cortex. We find that including traditionally omitted variables changes estimates of the original parameters and that modulation originally

attributed to one variable is reduced after new variables are included. In GLMs describing single-neuron dynamics, we then demonstrate how postspike history effects can also be biased by omitted variables. Here we find that omitted variable bias can lead to mistaken conclusions about the stability of single-neuron firing. Omitted variable bias can appear in any model with confounders-where omitted variables modulate neural activity and the effects of the omitted variables covary with the included effects. Understanding how and to what extent omitted variable bias affects parameter estimates is likely to be important for interpreting the parameters and predictions of many neural encoding models.

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95. Tensor Representation of Topographically Organized Semantic Spaces

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Human brains seem to represent categories of objects and actions as locations in a continuous semantic space across the cortical surface that reflects the similarity among categories. This vision of the semantic organization of information in the brain, suggested by recent experimental findings, is in harmony with the well-known topographically organized somatotopic, retinotopic, and tonotopic maps in the cerebral cortex. Here we show that these topographies can be operationally represented with context-dependent associative memories. In these models, the input vectors and, eventually also, the associated output vectors are multiplied by context vectors via the Kronecker tensor product, which allows a spatial organization of memories. Input and output tensor contexts localize matrices of semantic categories into a neural layer or slice and, at the same time, direct the flow of information arriving at the layer to a specific address, and then forward the output information toward the corresponding targets. Given a neural topographic pattern, the tensor representation will place a set of associative matrix memories within a topographic regionalized host matrix in such way that they reproduce the empirical pattern of patches in the actual neural layer. Progressive approximations to this goal are accomplished by avoiding excessive overlap of memories or the existence of empty regions within the host matrix.

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96. Multi-Instance Dimensionality Reduction via Sparsity and Orthogonality

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We study a multi-instance (MI) learning dimensionality-reduction algorithm through sparsity and orthogonality, which is especially useful for high-dimensional MI data sets. We develop a novel algorithm to handle both sparsity and orthogonality constraints that existing methods do not handle well simultaneously. Our main idea is to formulate an optimization problem where the sparse term appears in the objective function and the orthogonality term is formed as a constraint. The resulting optimization problem can be solved by using approximate augmented Lagrangian iterations as the outer loop and inertial proximal alternating linearized minimization (iPALM) iterations as the inner loop. The main advantage of this method is that both sparsity and orthogonality can be satisfied in the proposed algorithm. We show the global convergence of the proposed iterative algorithm. We also demonstrate that the proposed algorithm can achieve high sparsity and orthogonality requirements, which are very important for dimensionality reduction. Experimental results on both synthetic and real data sets show that the proposed algorithm can obtain learning performance comparable to that of other tested MI learning algorithms.

97. Use of a Deep Belief Network for Small High-Level Abstraction Data Sets Using Artificial Intelligence with Rule Extraction

Yoichi Hayashi

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We describe a simple method to transfer from weights in deep neural networks (NNs) trained by a deep belief network (DBN) to weights in a backpropagation NN (BPNN) in the recursive-rule eXtraction (Re-RX) algorithm with J48graft (Re-RX with J48graft) and propose a new method to extract accurate and interpretable classification rules for rating category data sets. We apply this method to the Wisconsin Breast Cancer Data Set (WBCD), the Mammographic Mass Data Set, and the Dermatology Dataset, which are small, high-abstraction data sets with prior knowledge. After training these three data sets, our proposed rule extraction method was able to extract accurate and concise rules for deep NNs trained by a DBN. These results suggest that our proposed method could help fill the gap between the very high learning capability of DBNs and the very high interpretability of rule extraction algorithms such as Re-RX with J48graft.

98. Limitations of Proposed Signatures of Bayesian Confidence

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The Bayesian model of confidence posits that confidence reflects the observer's posterior probability that the decision is correct. Hangya, Sanders, and Kepecs (2016) have proposed that researchers can test the Bayesian model by deriving qualitative signatures of Bayesian confidence (i.e., patterns that one would expect to see if an observer were Bayesian) and looking for those signatures in human or animal data. We examine two proposed signatures, showing that their derivations contain hidden assumptions that limit their applicability and that they are neither necessary nor sufficient conditions for Bayesian confidence. One signature is an average confidence of 0.75 on trials with neutral evidence. This signature holds only when class-conditioned stimulus distributions do not overlap and when internal noise is very low. Another signature is that as stimulus magnitude increases, confidence increases on correct trials but decreases on incorrect trials. This divergence signature holds only when stimulus distributions do not overlap or when noise is high. Navajas et al. (2017) have proposed an alternative form of this signature; we find no indication that this alternative form is expected under Bayesian confidence. Our observations give us pause about the usefulness of the qualitative signatures of Bayesian confidence. To determine the nature of the computations underlying confidence reports, there may be no shortcut to quantitative model comparison.

99. Bayesian Modeling of Motion Perception Using Dynamical Stochastic Textures

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A common practice to account for psychophysical biases in vision is to frame them as consequences of a dynamic process relying on optimal inference with respect to a generative model. The study presented here details the complete formulation of such a generative model intended to probe visual motion perception with

a dynamic texture model. It is derived in a set of axiomatic steps constrained by biological plausibility. We extend previous contributions by detailing three equivalent formulations of this texture model. First, the composite dynamic textures are constructed by the random aggregation of warped patterns, which can be viewed as three-dimensional gaussian fields. Second, these textures are cast as solutions to a stochastic partial differential equation (sPDE). This essential step enables real-time, on-the-fly texture synthesis using time-discretized autoregressive processes. It also allows for the derivation of a local motion-energy model, which corresponds to the log likelihood of the probability density. The log likelihoods are essential for the construction of a Bayesian inference framework. We use the dynamic texture model to psychophysically probe speed perception in humans using zoom-like changes in the spatial frequency content of the stimulus. The human data replicate previous findings showing perceived speed to be positively biased by spatial frequency increments. A Bayesian observer who combines a gaussian likelihood centered at the true speed and a spatial frequency dependent width with a "slow-speed prior" successfully accounts for the perceptual bias. More precisely, the bias arises from a decrease in the observer's likelihood width estimated from the experiments as the spatial frequency increases. Such a trend is compatible with the trend of the dynamic texture likelihood width.